

CARDIOVASCULAR DISEASE DETECTION USING IMAGE PROCESSING AND MACHINE LEARNING

*A Dissertation submitted in partial fulfillment of the requirements for the
award of the Degree of*

**BACHELOR OF ENGINEERING
IN
ELECTRONICS AND COMMUNICATION ENGINEERING**

BY

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Muffakham Jah College of Engineering and Technology

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MUFFAKHAM JAH
COLLEGE OF ENGINEERING & TECHNOLOGY
(Estd. by Sultan-Ul-Uloom Education Society in 1980)
(Affiliated to Osmania University, Hyderabad)
Approved by the AICTE & Accredited by NAAC

CERTIFICATE

This is to certify that the dissertation titled '**CARDIOVASCULAR DISEASE DETECTION USING IMAGE PROCESSING AND MACHINE LEARNING**' submitted by **Mohammed Ilyas Ahmed (Roll No: 1604-20-735-076)**, **Mohammed Hassan Ahmed (Roll No: 1604-20-735-091)**, and **Mirza Sameeullah Baig (Roll No: 1604-20-735-101)** in partial fulfillment of the requirements for the award of the Degree of **Bachelor of Engineering**, is a bonafide record of work carried out by them under my guidance and supervision during the academic year 2023-2024.

The results embodied in this thesis have not been submitted to any University or Institute for the award of any Degree or Diploma.

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DECLARATION

We hereby declare that the work presented in this dissertation entitled '**Cardiovascular Disease Detection Using Image Processing And Machine Learning**' submitted in partial fulfillment of the requirement for the award of the Degree of **Bachelor of Engineering**, in the Department of Electronics and Communication Engineering, Muffakham Jah College of Engineering and Technology, Hyderabad is an authentic record of our own work carried out from October 2023 to May 2024 under the guidance and supervision of Dr. Nazeerunnisa, Senior Assistant Professor, Department of ECE, MJCET.

We have not submitted the matter embodied in this dissertation for the award of any other degree or diploma.

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Mohammed Ilyas Ahmed

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ABSTRACT

The prevalence of Cardiovascular Diseases (CVD) currently poses a significant global burden to the healthcare systems. Arrhythmia is a CVD condition that can lead to death in chronic CVD patients or even to an apparent healthy person. Electrocardiogram (ECG) is the first-point diagnosis tool of arrhythmia. Currently, Implantable Cardioverter Defibrillator (ICD) and its subcutaneous version are used for treating fatal arrhythmias. However, these devices can only detect arrhythmia once it started and therefore in the case of fatal arrhythmias such as Ventricular Fibrillation they are not very effective in saving lives. Recently, ECG has been applied to predict impending arrhythmias but needed to be improved.

This study aims to develop a machine learning-based model for predicting the heart status of patients using electrocardiogram signals. Machine learning algorithms, such as support vector machines, random forests, and neural networks, were used to analyze ECG signals and predict heart status.

In this project, a deep learning-based approach for predicting heart status using electrocardiogram signals is used. Specifically, a convolutional neural network was employed to learn the complex features in ECG signals and predict the presence or absence of cardiac abnormalities. A large dataset of ECG signals from patients with normal and abnormal heart conditions is utilized to classify 12 rhythm classes using 91,232 single-lead ECGs from 53,549 patients who used a single-lead ambulatory ECG monitoring device. It proposes a robust and efficient 12-layer deep one-dimensional CNN on classifying the five micro-classes of heartbeat types in the MIT- BIH Arrhythmia database. The five types of heartbeat features are classified, and wavelet self-adaptive threshold denoising method is used in the experiments. These findings demonstrate that an end-to-end deep learning approach can classify a broad range of distinct arrhythmias from single-lead ECGs with high diagnostic performance similar to that of cardiologists. If confirmed in clinical settings, this approach could reduce the rate of misdiagnosed computerized ECG interpretations and improve the efficiency of expert human ECG interpretation by accurately prioritizing the most urgent conditions.

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LIST OF ABBREVIATIONS

AFL	Atrial Flutter
CVD	Cardiovascular Diseases
CNN	Convolutional Neural Network
DNN	Deep Neural Network
ECG	Electrocardiogram
MIT-BIH	Massachusetts Institute of Technology and Beth Israel Hospital
PVC	Premature Ventricular Contractions
RNN	Recurrent Neural Network
ReLU	Rectified Linear Activation Unit
RO	Ring Oscillator
SVM	Support Vector Machine
TEG	Thermo Electric Generator
NHB	Normal Heartbeat
AHB	Abnormal Heartbeat

LIST OF SYMBOLS

F	Frequency
H	Hertz
I	Current
LI	Lead I
L II	Lead II
L III	Lead III
V	Voltage
QRS	QRS Waveform
Ms	Millisecond

CHAPTER 1

INTRODUCTION

1.1 Introduction

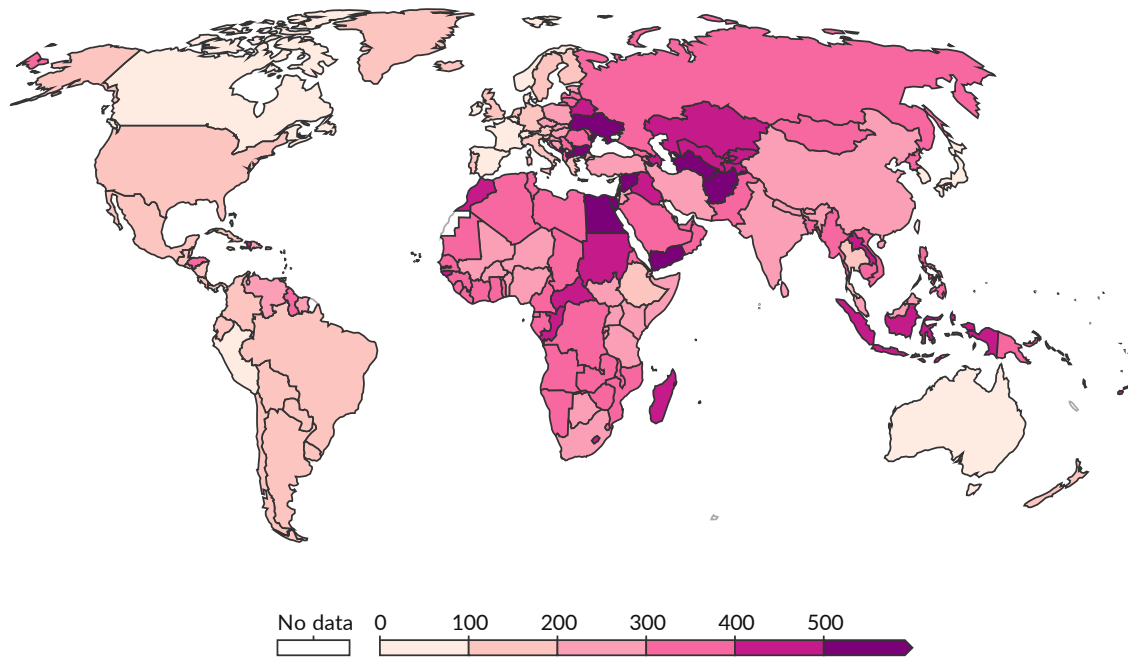
The medical field is expanding at a rapid pace as new diseases emerge on a regular basis, necessitating the development of appropriate treatment options. For appropriate diagnosis and treatment, an accurate and efficient mode of operation is required. If the scheme is automated, it could be extremely useful. However, the problem is that, medical practitioners are not effective enough in each subspecialty, resulting in a scarcity of resource people. As a result, implementing an effective automatic medical diagnostic scheme could be highly beneficial to all stratification involved in this process.

The heart is a muscular organ that is in charge of blood circulation. The heart pumps deoxygenated blood from other areas of the body into veins, as well as oxygenated blood is pumped back from the lungs to different areas via arteries. Lungs help in the process of oxygenation. The control the contraction and relaxation of the atrium and ventricles are actuated via messages sent over the heart muscle tissues defined.

As per to WHO statistics, heart diseases claim the lives of 12,000,000 people worldwide every year. CVD report for nearly 1 2 of deaths in the US or another advanced nations, and the same could be said for creating countries such as India. As a result, cardiac disorder are the leading source of death in adults worldwide. As per [3], a recent cardiac health-related survey, nearly 1.2 billion population die every year as a outcome of heart diseases. There is no single solution to the increasing load of HD. There is no single solution to the rising burden of heart disease, given the massive transformation in ethnic, as well as economic environments. Heart failure prognosis has historically been an extremely thought-provoking task in the eve of high-cost ratios. The price of a wide range of modern imaging and clinical methodologies for the diagnosis of heart disease is prohibitively high. Leading causes of cardiac disease involve chest discomfort, dyspnoea, fatigue, edema, palpations, as well as syncope, as well as cough, haemoptysis, and cyanosis. Figure 1.1 depicts the 2017 heart disease death rate framework.

Death rate from cardiovascular diseases, 2021

Estimated annual death rate from cardiovascular disease¹ per 100,000 people.



Data source: IHME, Global Burden of Disease (2024)

OurWorldInData.org/causes-of-death | CC BY

Note: To allow for comparisons between countries and over time, this metric is age-standardized².

1. Cardiovascular disease: Cardiovascular diseases cover all diseases of the heart and blood vessels – including heart attacks and strokes, atherosclerosis, ischemic heart disease, hypertensive diseases, cardiomyopathy, rheumatic heart disease, and more. They tend to develop gradually with age, especially when people have risk factors like high blood pressure, smoking, alcohol use, poor diet, and air pollution.

2. Age standardization: Age standardization is an adjustment that makes it possible to compare populations with different age structures by standardizing them to a common reference population. [Read more: How does age standardization make health metrics comparable?](#)

Figure. 1.1: Global Map Standardized Death Rate of CVD in 2021

This chapter provides an introduction to cardiovascular diseases and their statistics in medical record. The cardiac cycle is a series of pressure changes that take place within the heart. These pressure changes result in the movement of blood through different chambers of the heart and the body as a whole. An understanding of the cardiac cycle is vital to the interpretation of cardiac imaging. In particular, one of the most direct methods for cardiac imaging is the electrocardiogram. The electrocardiogram is a visual time series that records the electrical activity generated by each cardiac cycle as shown in Figure 1.2. This non-invasive detection method is easy to operate and has become an essential tool for assisting doctors in analysing pathology.

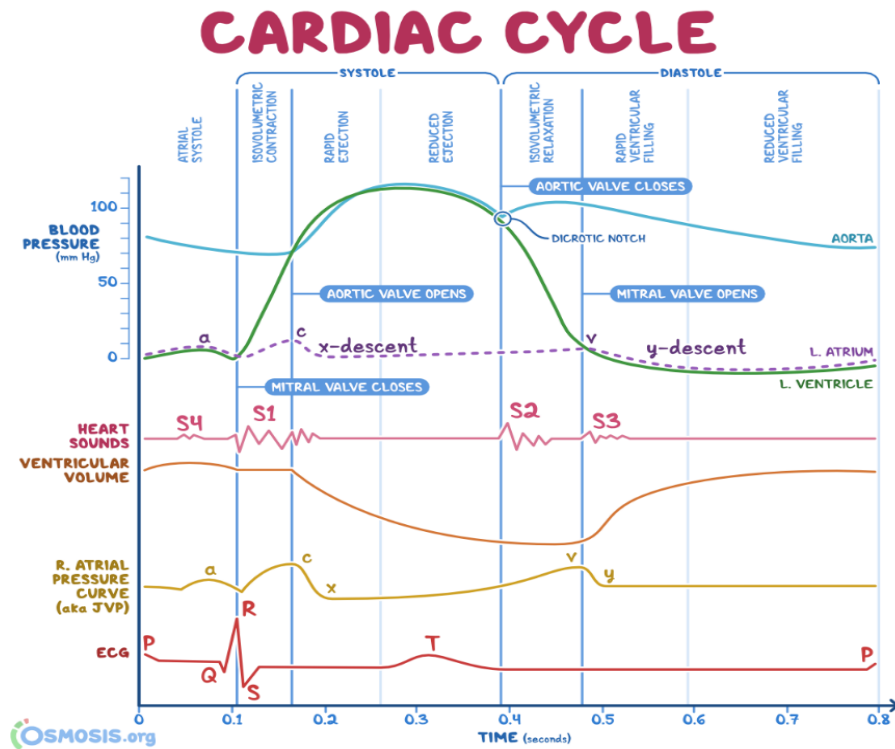


Figure. 1.2: Cardiac Cycle

1.2 Current Scope

The current scope of using machine learning models for predicting heart status with electrocardiogram (ECG) signals is quite broad and rapidly expanding. Some of the current areas of research and development in this field include:

1. Developing new machine learning models: Researchers are continually developing new machine learning models that can effectively analyse ECG signals and accurately predict heart status. These models include deep learning models, such as convolutional neural networks and recurrent neural networks , as well as traditional machine learning models, such as support vector machines (SVMs) as per [3] random forests.
2. Improving the accuracy of machine learning models: Improving the accuracy of machine learning models is a key area of research in this field. Researchers are exploring new techniques for data pre-processing, feature extraction, and model optimization to improve the accuracy of heart status prediction.

3. Developing explainable machine learning models: The lack of interpretability of machine learning models is a major challenge in healthcare. Therefore, researchers are developing new techniques to make the model's predictions more interpretable and transparent for clinicians.

4. Real-time monitoring: Real-time monitoring of ECG signals is crucial for early diagnosis and treatment of cardiac diseases. Therefore, researchers are developing machine learning models that can be deployed on wearable devices [8] for real-time monitoring and prediction of heart status.

Overall, the current scope of using machine learning models for predicting heart status with ECG signals is quite broad, as shown in figure 1.3 and ongoing research in this area has the potential to significantly improve the accuracy and efficiency of cardiac diagnosis and monitoring.

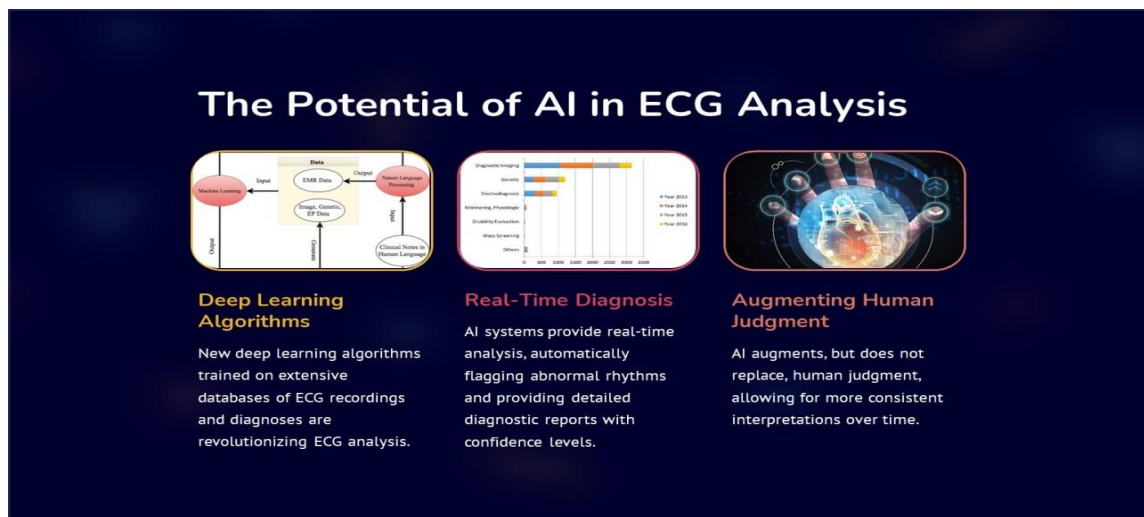


Figure 1.3 Scope of Machine Learning in ECG analysis

1.3 Problem Statement

Early detection of arrhythmias before major heart failures can reduce cardiac arrests in patients by suitable numbers.

The problem addressed in this project is the accurate prediction of heart status using electrocardiogram signals, which is essential for early diagnosis and treatment of cardiac diseases. While ECG signals are commonly used in clinical settings for cardiac diagnosis, their interpretation is a complex and time-consuming process. Machine learning-based approaches, such as convolutional neural networks, have shown promise in automating the analysis of ECG signals and predicting heart

status. However, there is a need for further research to improve the accuracy and efficiency of CNN-based models for heart status prediction, especially in real-world clinical scenarios.

1.4 Motivation

Predicting heart status with electrocardiogram signals using machine learning models stems from the need to improve the accuracy and efficiency of cardiac diagnosis and monitoring. Cardiac diseases are a leading cause of death worldwide, and early diagnosis and treatment are critical for improving patient outcomes. ECG signals contain valuable information about the electrical activity of the heart and are widely used in clinical settings for cardiac diagnosis and monitoring. However, interpreting ECG signals is a complex and time-consuming process that requires extensive training and expertise. Additionally, the development of wearable devices and mobile health technologies has increased the demand for real-time monitoring of ECG signals, which can be facilitated by machine learning models. Overall, the motivation for predicting heart status with ECG signals using machine learning models is to improve the accuracy, efficiency, and accessibility of cardiac diagnosis and monitoring, ultimately leading to better patient outcomes.

1.5 Objectives

The objective of predicting heart status with electrocardiogram signals using convolutional neural networks is to develop an accurate and efficient machine learning-based approach for automated cardiac diagnosis and monitoring. Specifically, the objectives of using CNNs for heart status prediction with ECG signals include:

1. To develop a CNN-based model that can accurately classify ECG signals into normal and abnormal heart conditions, thereby improving the healthcare sector with advanced technology.
2. To evaluate the performance of the CNN-based model using a large dataset of ECG signals from patients with both normal and abnormal heart conditions.
3. To Compare the performance of the CNN-based model with other commonly used machine learning models for ECG analysis, such as support vector machines and random forests.
4. To Investigate the sensitivity and specificity of the CNN-based model, as well as its ability to generalize across different populations and datasets.

5. To Explore techniques for improving the interpretability of the CNN-based model, facilitating clinical decision-making and enhancing patient outcomes.

1.6 Applications

The prediction of heart status with electrocardiogram signals using convolutional neural networks (CNNs) has several potential applications in healthcare. Some of these applications include:

1. Early detection of cardiac diseases
2. Personalized or self cardiac monitoring
3. To help doctors with research
4. Decision support for clinicians
5. Wearable electronic health monitoring devices

1.7 Organization of the Thesis

Chapter 1: Is the introduction to the thesis. It consist of problem statement, motivation followed by objective and applications.

Chapter 2: This provides brief overview of all the literature works reviewed.

Chapter 3: Introduces to machine learning and its algorithms.

Chapter 4: Discusses about convolutional neural network in detail.

Chapter 5: Step by step approach using MIT-BIH dataset.

Chapter 6: Various charge pumps are discussed and the reason behind why they were chosen is provided.

Chapter 7: Gives relevant results for different csv files.

Chapter 8: Provides conclusion and future scope for this work.

Chapter 9: Provides references for the same

CHAPTER 2

LITERATURE SURVEY

2.1 Introduction

Various existing literature works in the areas concerned with this project have been reviewed to get a clear and better understanding of the different aspects involved in the project. The titles of these literature works have been listed in the reference section in the thesis. This chapter presents a brief overview of all the literature works that were reviewed.

2.2 Synopsis of the Previous Work

Some of the key findings from previous studies are:

1. In this paper[1] a method for ECG heartbeat classification based on a transferable representation. Specifically, a trained deep convolutional neural network with residual connections for the arrhythmia classification task and shown that the representation learned for this task can be used as a base to train accurate classifiers for the classification of MI. According to the results, the suggested method is able to make predictions on both tasks with accuracies comparable to the state of the art methods in the literature. Furthermore, they have visualized the learned representation using t-SNE method and illustrated the effectiveness of the proposed approach.
2. In this paper [2] ,use of a proposed EEMD-based ECG feature extraction method for human identification was evaluated. A robust preprocessing scheme was used compensate for noise, HRV, and low-quality ECG signals. EEMD and Welch spectral analysis were applied to each single-lead ECG signal to extract morphological and spectral information. These two kinds of features were used and their dimensionality reduced using the PCA method. The system was validated using 45 subjects from three public ECG databases at a variety of heart rates, and the results demonstrate the feasibility of the proposed method as a biometric system. The proposed method performed ideally achieving a high recognition rate.
3. In summary,[3] This work has validated an ability to classify heartbeats. Classification process is using some features of heartbeats and machine learning classification algorithms with local host pc working using only one node, which are crucial for diagnosis of cardiac arrhythmia. The developed GBT and RF models can classify different ECG heartbeat types and thus, can be implemented into a

CAD ECG system to perform a quick and reliable diagnosis. The proposed model has the potential to be introduced into clinical settings as a helpful tool to aid the cardiologists in the reading of ECG heart-beat signals and to understand more about them. The occurrence, sequential patterns and persistence of the classes of ECG heartbeats considered in this work can be grouped under three main categories which represents normal, PVC, PAC, and other.

4. In this paper, a deep neural network(DNN), for automatic classification of ECG signals was used differentiated from the normal ones. DNN has been developed using tensor flow framework.

5. This paper [4] presents An experimental platform is constructed for ECG acquisition to collect ECG signals for arrhythmia classification to verify the effectiveness of the proposed method for ECG identification. We utilize the FLUKE ProSim 2 vital sign simulator as the signal resource to provide six types of ECG signals, and we select the standard I lead ECG in this platform. During ECG signal acquisition, an ADuCM361 microcontroller is used to control the A/D converter (Analog Devices) for converting analogue ECG signals into digital signals. The acquired ECG signals are transmitted via a USB cable to a PC and are processed by the proposed method for classification. Moreover, the ECG acquisition module is supplied with power via the USB cable.

6. In this work [6], A pre-processing step was performed on the electrocardiograms (ECG) for baseline removal. Next, a QRS complex detection algorithm was implemented to detect the heartbeats, which contain the primary information that is employed in the classification approach. Next, ECG segmentation was performed, by which a set of features based on the RR interval and the beat waveform morphology were extracted from the ECG signal. The size of the feature vector was reduced by principal component analysis. Finally, the reduced feature vector was employed as the input to an artificial neural network.

7. In this paper [7], The experience of the past 20 years since the publication of the MIT-BIH Arrhythmia Database, and the AHA Data-base shortly thereafter, can be regarded as a grand experiment in shaping the direction of development of arrhythmia detectors. Until the databases became available, performance statistics were of little or no value since it was widely understood that each manufacturer designed its products using its own data, and designed its statistics to present the product in a favourable light. Conscientious developers who sought to make their algorithms more accurate faced pressure to match their competitors' products feature for feature, rather than spending

effort and money making improvements that could not be quantified, and therefore added and perceived value to the product.

8. In this paper [8], Hidden Markov Models (HMMs) have been used with great effect in classifying ECGs. Andreadou et al was able to demonstrate an accuracy of 99.97% in detecting the QRS complex of the heartbeat. Their approach was to create a general model of the heartbeat, and then tune the model to each individual by using data from the first 20 seconds of their ECG. The general model of the heartbeat was composed of discrete states representing the P, Q, R, S, and T waves, the PQ and ST intervals, and the isoline. Andreadou et al's work was able to detect premature ventricular contractions .

9. In this research paper [9], Computerized electrocardiogram interpretation plays a critical role in the clinical ECG workflow. Widely available digital ECG data and the algorithmic paradigm of deep learning present an opportunity to substantially improve the accuracy and scalability of automated ECG analysis. However, a comprehensive evaluation of an end-to-end deep learning approach for ECG analysis across a wide variety of diagnostic classes has not been previously reported. Here, we develop a deep neural network to classify 12 rhythm classes using 91,232 single-lead ECGs from 53,549 patients who used a single-lead ambulatory ECG monitoring device. When validated against an independent test dataset annotated by a consensus committee of board-certified practicing cardiologists, the DNN achieved an average area under the receiver operating characteristic curve (ROC) of 0.97.

10. In this paper [10], an algorithm for predicting VA through step-by-step analysis of ECG signal coupled with innovative signal processing and machine learning methods is presented. We have first introduced an automated ECG delineation algorithm with high precision. The experimental results in terms of positive detection (+P) and sensitivity (SE) that tested on MIT-BIH database and QT database are used to evaluate the proposed algorithm. The results also compared with other formally published methods and show a high performance for ECG feature extraction. Our algorithm achieved 99.89% of Se and 99.94% of +P, 100% of Se and 99.83% of +P for R peak detection in MIT-BIH and QT database respectively. In the UHS analysis we demonstrate a sensitivity of 99.88% and a positive predictive value of 99.99% with reductions in TWOS episodes (from 166 to 0). Whilst in the MIT-BIH analysis we demonstrate a sensitivity of 99.87% and a positive predictive value of 99.99%, with effective reduction in the number of false positives in R wave detection. We have proposed a novel algorithm based on the combination of PSR and FCM clustering technique of the ECG for prediction and classification of VA. 32 healthy subjects and 32 arrhythmic subjects from PTBDB and CUDDB

respectively were used to evaluate our proposed technique. The proposed system showed 298.3 seconds of an average prediction time for impending ventricular arrhythmia based on 32 VA subjects. Also, our system can classify four types of VA (VA without VPBs, VF, VT, and VT followed by VF) with an average 238.3 seconds before the VA onset. It achieved an average accuracy of 98.4%, a sensitivity of 97.5% and specificity of 99.1%, which was comparable and better than the previous works. This study has demonstrated that this algorithm can be used to predict and classify the impending VA. Finally, a simple PSR-based arrhythmia classifier has been developed to classify normal ECG, atrial fibrillation, VT, VF and VT followed by VF. This process was verified with 122 records with more than 5500 windows of ECG signals. The results show an average sensitivity of 98.73%, specificity of 99.71% and accuracy of 99.56%.

2.3 Summary

Sl.No	Title	Authors	Conferences	Methods	Remarks
1.	Heartbeat classification system based on neural networks and dimensionality	Dalvi Rdaf, Zago GT, Andreão RV	Research on Biomedical Engineering 2022	Neural network and SVM were used for classification.	Performance on a test set of only 18 ECG records of 30 min each achieved an accuracy of 96.97%.
2.	A generic and robust system for automated patient-specific classification of ECG signals.	Ince T, Kiranyaz S, Gabbouj M.	IEEE Trans Biomed Eng-2020	Feed-forward and fully connected artificial neural networks aided by particle swarm optimization.	Two patterns of heartbeats are recognized with 96% accuracy.
3.	Using hidden Markov models and spark to mine ECG data	O'Brien J.	Saint Mary's College of Maryland Journal of 2019	Combining accurate Hidden Markov models (HMM) techniques with Apache Spark to improve the speed of ECG analysis.	The paper has proven that there is potential for developing a fast classifier for heartbeats classification
4.	Big data analytics in genomics: the point on deep learning solutions.	Celesti F, Celesti A, Carnevale L, Galletta A, Campo S, Romano A, Bramanti P, Villari M. B	2019 IEEE symposium on computers and communications (ISCC)	DNN (Deep Neural Network) has been used for deep learning to classify heartbeats.	classification was only two types (Normal and Abnormal) and dataset size was almost 85,000 records.

CHAPTER 3

Machine Learning from Theory to Algorithms

3.1 Introduction

This chapter describes about machine learning, its implementation and approaches. It deals with different stages of machine learning. Here, the different models are discussed along with their algorithms .

3.2 Machine Learning

Machine learning is a multi-disciplinary field having a wide-range of research domains reinforcing its existence. These are as shown in the following Figure. The simulation of ML models is significantly related to Computational Statistics whose main aim is to focus on making predictions via computers. It is also co-related to Mathematical Optimization as which relates models, applications and frameworks to the field of statistics. Real world problems have high complexity which make them excellent candidates for application of ML. Machine learning can be applied to various areas of computing to design and program explicit algorithms with high performance output, for example, email spam filtering, fraud detection on social network, online stock trading, face & shape detection, medical diagnosis, traffic prediction, character recognition and product recommendation among others. The self-driving Google cars, Netflix showcasing the movies and shows a person might like, online recommendation engines like friend suggestions on Facebook, “more items to consider” and “get yourself a little something” on Amazon, and credit card fraud detection, are all real-world examples of application of machine learning [6]. Machine learning methods are critical in data analysis as well as prediction.

Traditional methods fail to deal with medical data extraction because it is enormous as well as complicated. The primary motive of the ML case research is to form a recommendation system for heart disease diagnosis or present novel & high opportunity methods for early treatment of heart disease diagnosis. Developers focus on heart disease databases in specific as well as look at apps in chronic disease management [7]. In everyday life, machine learning is used in critical decision-making structures like medical diagnosis and clinical decision-making.

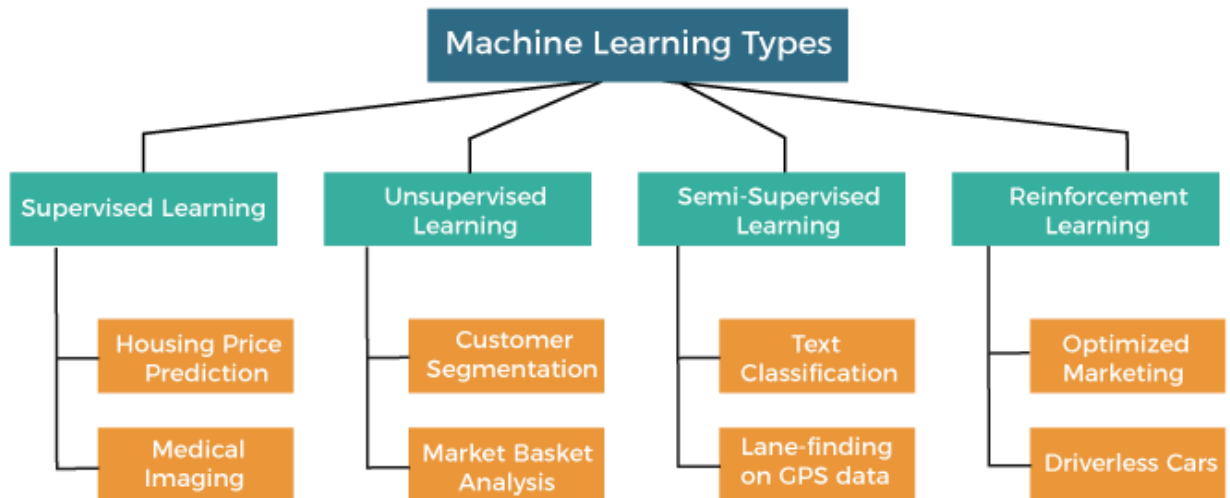


Figure 3.1: Classification on Machine Learning Techniques

Figure 3.1 above illustrates machine learning algorithm for different uses. There are many different kinds of machine learning methodologies, but they are widely divided into four categories based on their objective:

1.Supervised learning: In this learning, a feature is implied from labeled training data that defines an incoming information to an output based on the function learned from a set of labeled training instances as a shown in figure3.2.

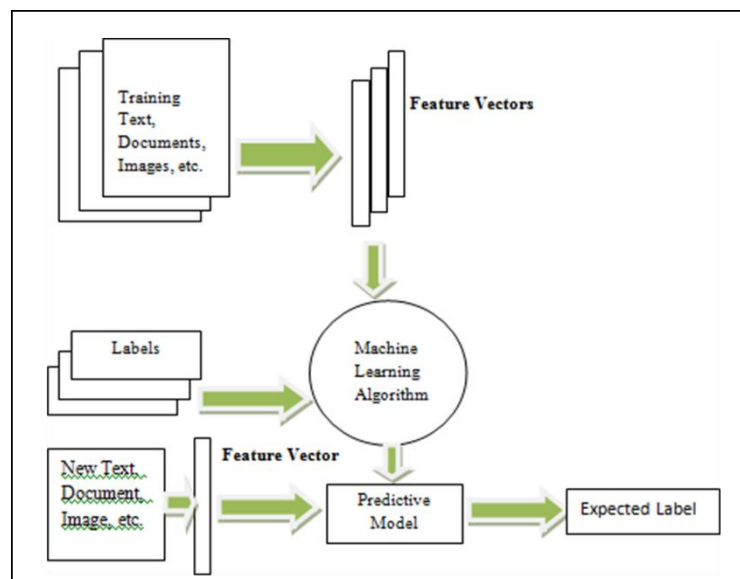


Figure 3.2: Flow Chart Of Supervised Learning

2. Unsupervised learning: Due to the unlabeled dataset, prior training is not provided in this aspect of teaching. As shown in figure 3.3 The machine is limited to grouping unsorted information by itself based on similarities, differences, as well as hidden patterns in the data .

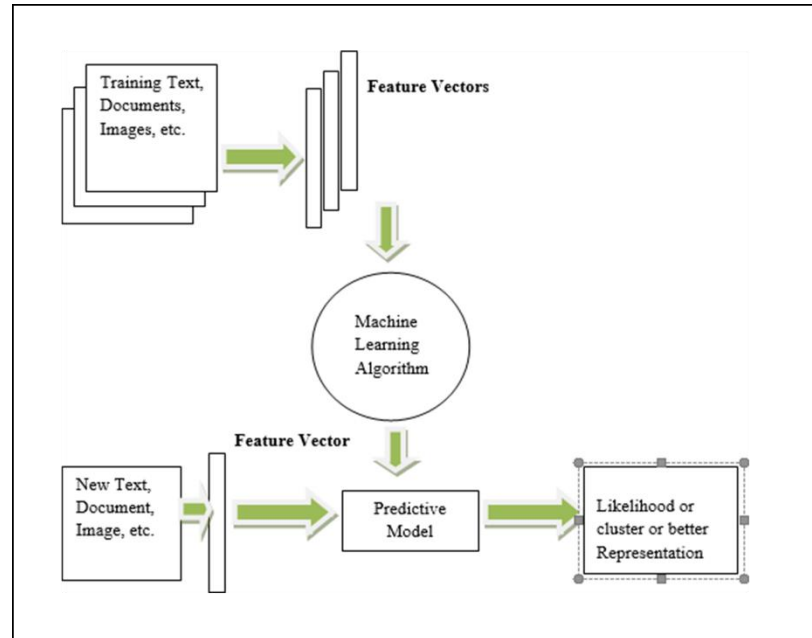


Figure 3.3: Flow chart Of Unsupervised Learning

3. Semi-supervised learning: It drops somewhere among supervised as well as unsupervised learning because the input information in semi-supervised learning is partly labeled .

4. Reinforcement learning: A machine is trained using the trial-and-error technique in this learning. The algorithm learns from its previous experiences until it has explored all possible states and has determined an ideal behavior to achieve optimal performance. It's commonly utilized in robotics, gaming, as well as navigation.

3.3 Classification Of Different Machine Learning Techniques

Machine Learning defines the numerous types of approaches that can be seen in figure 3.4 used in different fields. SVM, LR, decision trees, RF , K-nearest neighbor (K-NN), naive bays (NB), neural networks (NN), and linear discriminate analysis (LDA) ,as per [9], are some of the most widely used applications in solving supervised learning issues. SVM, LR, DT, RF , K-NN , nave bays, neural networks , and linear discriminate analysis (LDA) are some of the most widely used applications in solving supervised learning issues.

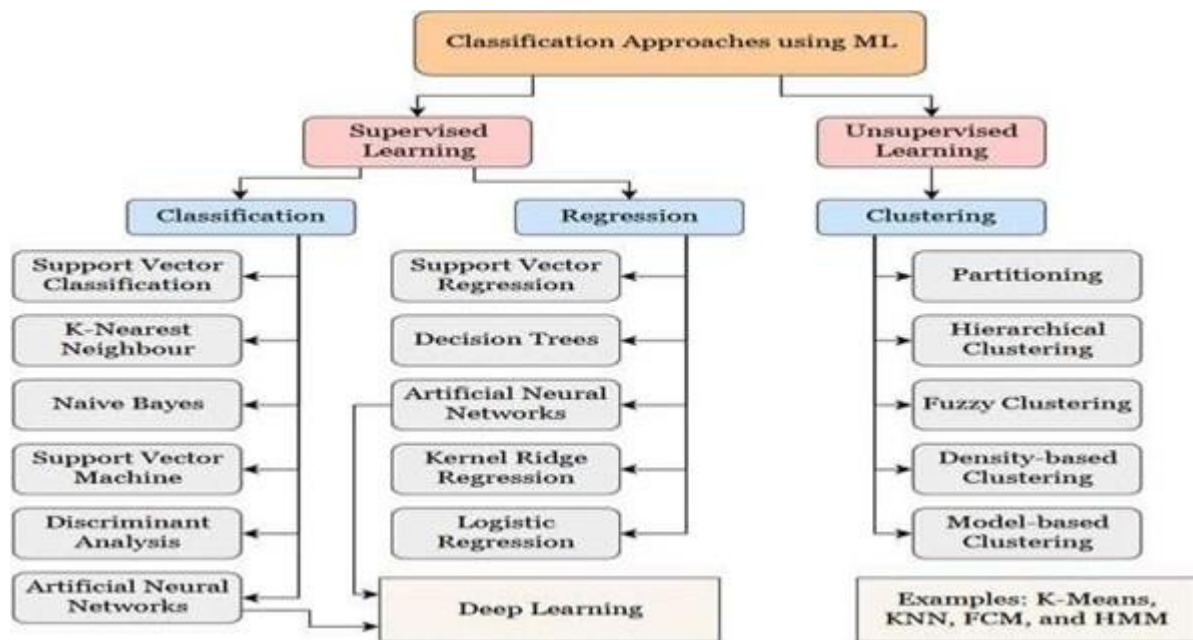


Figure 3.4: Classification of Machine learning techniques

1. Support Vector Machine (SVM) : The SVM approach [10] could be utilized for both regression as well as classification issues , but it is most commonly used to accomplish classification goals. By deciding a hyper-plane, the data is divided into classes. In way to select the best hyper-plane, the method aims at maximizing the distance among the data points of the classes. Margin maximization is a term used to describe the concept of distance maximization. There are 2 kinds of SVM algorithms: linear SVM as well as non-linear SVM. A hyper-plane is used to classify training data in linear SVM. A non-linear SVM is achieved by implementing a kernel trick to highest-margin hyper-planes, in which each pixel value is substitute via a NL kernel variable.

2. Decision Tree : It is a visual representation that uses branching technique to show all of the possible results of a decision based on some circumstances. The root node is the starting point in a decision tree, in which each internal node denotes a test a function, every tree branch reflects a decision rule, & every leaf node indicates an outcome (target variable). This method gives a graphical representation of a decision problem or aids in communication improvement and decision making under unstable situations. It also aids in decision making by enabling a data scientist to navigate forward & backward calculation routes as in [4].

In addition, decision trees are resistant to missing values or errors. The method is best suited for troubles with attribute-value pairs as instances.

The decision tree is categorized into two categories:

i) classification Tree, and ii) Regression Tree, The algorithm has applications in a variety of fields, including finance, banking, remote sensing, and medical centres

3. Random Forest (RF) : This method [12] creates a slew of decision trees from a random subset of data and is typically equipped using the bagging method .The classifier is constructed numerous times on random samples before making the final estimation by integrating the outcome of all decision trees to attain the optimum prediction .This method is resilient to outliers as well as maintains accuracy even missing data is present. It is extremely capable of handling binary, numerical, and categorical features without transformation or alteration. It also estimates the characteristics that are most important for classification. This method has recently been used in banks, the automobile industry, the healthcare industry, speech recognition, as well as the classification of images and texts.

4. Regression Models: Regression models are the most important component of the predictive analysis process. In overall, regression analysis use statistical methods to calculate the criterion of the distinction among forecasted and expected values. The regression design as per [16] makes use a variety of data types. It employs mathematical formulas to facilitate the design that depicts the relationship among the various parameters in the given database.

5.Clustering : In simple terms, it makes it easier for the user to determine the principles of the database. A good clustering technique will always have a high intra-class clustering and a low interclass clustering. The accuracy with which hidden patterns linked with information are clustered. Stream analysis, economic science, and several other fields make extensive use of clustering methods.

6. K-nearest neighbors algorithm(K-NN): K-NN used in [11] is a non-parametric technique, which means it makes no assumptions about data distribution The methodology could be utilized to resolve both classification as well as regression issues. It is a straightforward method in which all obtainable cases are regarded, as well as forecasting on a specific case is conducted using similarity measures. The k-NN method measures similarity using Euclidian, Manhattan, Minkowski, and Hamming distance functions. It should be mentioned that the Hamming distance measure is only appropriate to categorical target variables, whereas the other three are accurate for continuous target variables.

3.4 Deep Learning Approach

DL is a new subfield of ML that was developed to resolve the shortcomings of conventional machine learning methods in the automation period. Because the quality of most classification systems based on traditional methods is dependent on the feature extraction stage. It is frequently hard and time-consuming to collect valuable features from information.

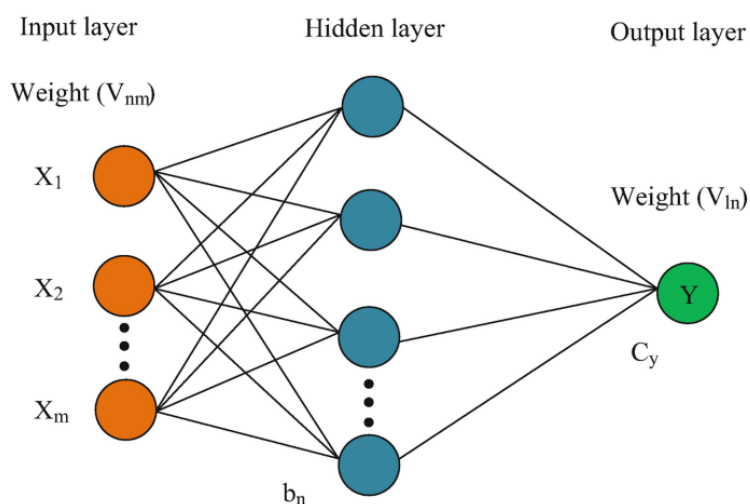


Figure 3.5 : Neural network

It has been used in a range of systems with remarkable results over the last five years[8]. The deep learning algorithm as shown in figure 3.6 attempts to learn without the use of any supervision. Deep Learning employs both supervised and unsupervised classification. Hidden layers learn features, which are then trained to develop a model.

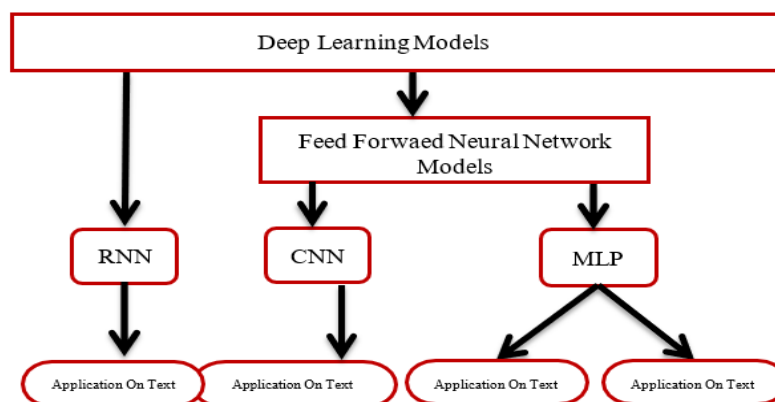


Figure 3.6: Deep learning Models

1.Feed Forward Neural Network

It designs as well as recursive neural network designs were used to classify DL techniques for sentiment classification. Feed-forward neural network models were divided into two types: multilayer perceptron models and convolutional neural network models. LSTM, a type of RNN is used in suggested model with some sort of Enhancement.

2.Convolutional Neural Network

CNNs are deep artificial NN that are primarily used to extract features, cluster focused on the similarities, as well as recognize objects. Similar methods are utilized to recognize faces, street signs, tumors, platypuses, as well as other elements of graphic data.

3.Recurrent Neural Network

RNNs are a kind of neural network in which the current inputs of hidden layers are determined by the previous outputs of hidden layers. This enables them to interact with a time sequence with temporal relations, like speech recognition as well as text classification, among other things. RNNs [17] are found to be more effective in sentiment analysis unlike CNNs in the literature. Consider the sentence or document D_i as an example. The document at hand is a sequence of things. Features are obtained from the words and represented in a high-dimensional vector space.

4.Long Short Term Memory (LSTM)

NN typically operate as a black box, making decisions based on inputs. It includes data about learning opportunities in static memory in the form of weights. The LSTM system was created to present explicit representation for memory in RNNs. The memory unit in the system is referred to as a 'cell,' but these designs are RNN adaptations that are suitable for sequential data.

3.5 The Generic Model of ML

ML is used to solve various problems that require learning on the part of the machine. A learning problem has three features:

- Task classes (The task to be learnt)
- Performance measure to be improved
- The process of gaining experience

For example, in a game of checkers, the learning problem can be defined as:

- Task T: Playing the game
- Performance Measure P: number of games won against the opponent.
- Experience E: practicing via playing games against itself and consistently improving the performance.

The generic model of machine learning consists of six components independent of the algorithm adopted. The following Figure 3.7 depicts these primary components.

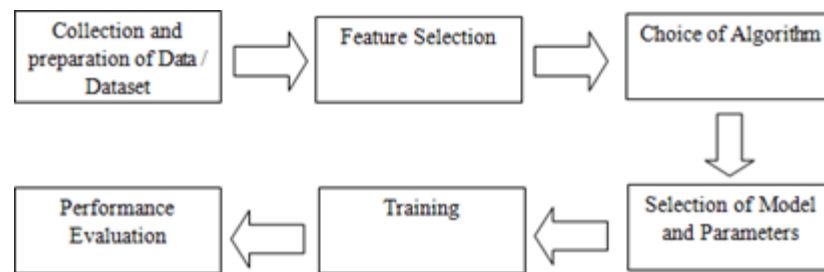


Figure 3.7 Components of a Generic ML mode

Each component of the above figure 3.7 has a specific task to accomplish as described next.

I. Collection and Preparation of Data: The primary task of in the machine learning process is to collect and prepare data in a format that can be given as input to the algorithm. A large amount may be available for any problem. Web data is usually unstructured and contains a lot of noise, i.e., irrelevant data as well as redundant data. Hence the data needs to be cleaned and pre-processed to a structured format.

II. Feature Selection: The data obtained from the above step may contain numerous features, not all of which would be relevant to the learning process. These features need to be removed and a subset of the most important features needs to be obtained.

III. Choice of Algorithm: Not all machine learning algorithms are meant for all problems. Certain algorithms are more suited to a particular class problem as explained in the previous section. Selecting the best machine learning algorithm for the problem at hand is imperative in getting the best possible results.

IV. Selection of Models and Parameters: Most of machine learning algorithms require some initial manual intervention for setting the most appropriate values of various parameters.

V. Training: After selecting the appropriate algorithm and suitable parameter values, the model needs to be trained using a part of the dataset as training data.

VI. Performance Evaluation: Before real-time implementation of the system, the model must be tested against unseen data to evaluate how much has been learnt using various performance parameters like accuracy, precision and recall.

Table 3.1 Applications of Machine Learning

Application	Description
Playing Checkers Game	A computer program learns to play checkers game, improves its performance as ability to win at various class of tasks involving the game, through experience obtained by playing games against itself.
Speech Recognition	Example: SPHINX system learns speaker specific sounds and words from speech signals.
Autonomous Vehicles	Example: Google Driver Less Cars , Tesla Cars. Machine Learning techniques are also highly effective in controlling sensor-based applications.
Filtering Email (Spam Email)	Machine learning helps to classify emails as spam emails by user. When new email arrives, machine learning will search, compare and based on previous spam emails by users. If new mail matches any of them they will be marked as same.
Robotics and Artificial Intelligence	Machine Learning is regarded as improvised approach to problem solving. using base knowledge and training with ml models, learning can be improved which will take robotics and AI to next generation levels.
Web and Social Media	1.Text mining, Spam filtering and Classifying Web page 2.Purchase Websites 3.Facebook uses Navie's Bayer's to analyze status update expressing positive and negative emotions.
Medical field	TRISS : Trauma and Injury Severity Score which is widely used to morality in injured patients using logistic regression.
Bayesian Methods	Bayer's Theorem is one of the most popular methods for calculating conditional probabilities given the set of hypothesis.

3.6 Conclusion

In conclusion, machine learning algorithms have shown tremendous potential in a wide range of applications, including image recognition, natural language processing, and predictive modelling. They are particularly useful in tasks where traditional rule-based approaches may not be feasible, and where large volumes of data are available.

Supervised learning algorithms, such as decision trees, random forests, support vector machines, and neural networks, are used for classification and regression tasks. Unsupervised learning algorithms, such as clustering and dimensionality reduction, are used for tasks such as pattern recognition, data visualization, and feature extraction.

Deep learning, a subfield of machine learning, has revolutionized the field of artificial intelligence, enabling the development of highly accurate and complex models for tasks such as image recognition, speech recognition, and natural language processing. Convolutional neural networks, recurrent neural networks, and generative adversarial networks are some of the popular deep learning models.

However, the performance of machine learning algorithms is highly dependent on the quality and quantity of data used for training. The ethical implications of machine learning algorithms, such as bias and fairness, are also important considerations that must be addressed in their development and deployment.

CHAPTER 4

Electrocardiogram (ECG) and Feature Extraction

4.1 Introduction

This chapter describes in detail about the human heart. A human heart is a muscular organ that pumps blood to the body. It is comprised of four chambers: the right and left atria, and the right and left ventricles - Figure shows an anterior view of the heart.

4.1.1 The Heart Structure

The main function of the heart is to circulate the blood. In every heartbeat, blood is squeezed into the arteries from the heart and then returns back to the heart in a one-way circuit through the veins. First the left atrium receives oxygenated blood from the lung via the pulmonary veins, and empties into the left ventricle through mitral valves. From there, the left ventricle injects the blood into the whole body via the aorta. The peripheral circulation system consumes the oxygen and nutrition, and then returns the blood with carbon dioxide as well as waste back to the right atrium via the superior vena cava. The right atrium then pulls the deoxygenated blood into the right ventricle through the tricuspid valve. The right ventricle then forces the blood through the semilunar valve and pulmonary arteries into the lung, where oxygen diffuses into the blood and is exchanged for carbon dioxide. The delivery of the oxygen and nutrients takes place in the connection between arteries and veins, known as the capillary network, which consisted of micro vessels.

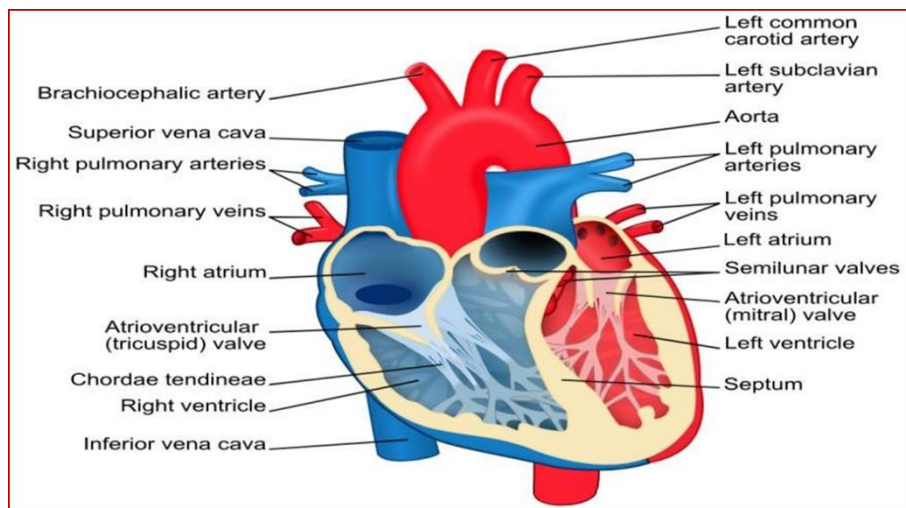


Figure 4.1: Structure of the heart

4.2 Electrocardiogram (ECG)

4.2.1 Components of the ECG

The heart activity can be considered as propagation of electrical impulses and therefore, any abnormality in heart function is reflected in this electrical impulse propagation characteristics. This could be measured by attaching electrodes to the surface of the body skin and recording the electrical activity by a device. Such recording is generally referred as Electrocardiogram (ECG) and it plays a significant role in day-to-day clinical practice for detecting arrhythmias [19] since the Cardiologists can obtain critical signs of malfunction of heart from ECG signal. In general, as shown in Figure 2.3, one normal ECG beat consists of the P-QRS-T and U waves and the relevant ECG features, such as PR interval, QT interval, QRS complex as well as ST segment and PR segment. In the following, a separate description of each of the components will be given to summarize their characteristics and the corresponding cardiac events.

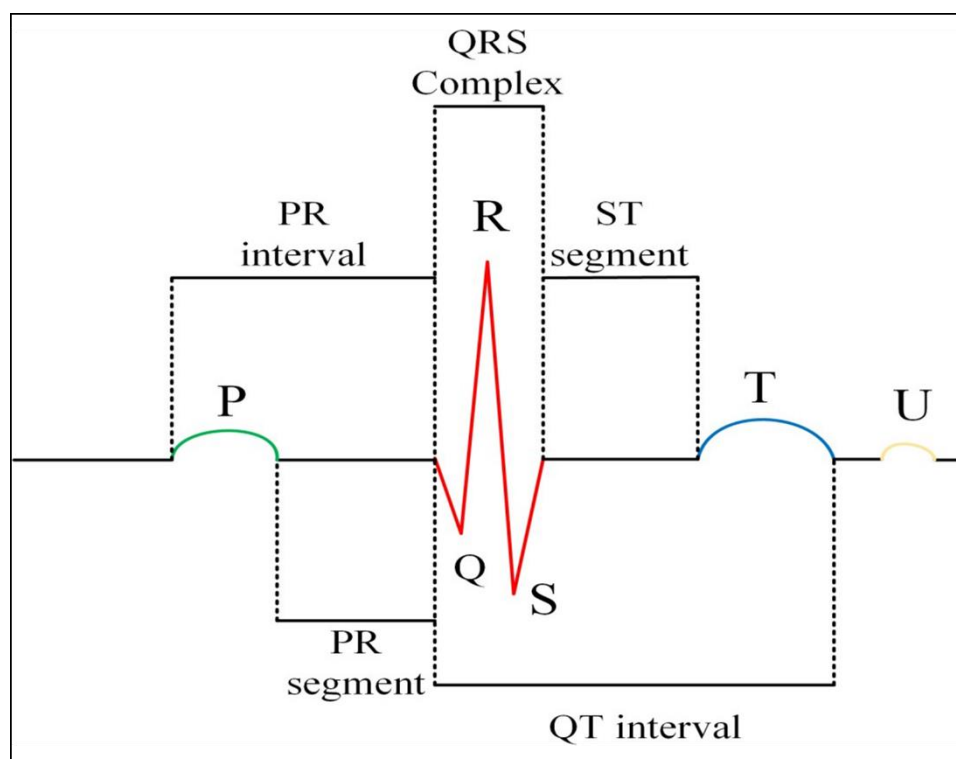


Figure 4.2: A normal ECG period

- **P wave:** The first wave in an ECG is called P wave which is generated due to electrical activity through the Atria. The normal duration of the P wave is shorter than 80 milliseconds (ms).

- **PR interval:** It is identified as the time for electrical impulse to travel from SA to AV node. The normal duration can vary between 120 and 200 ms.
- **PR segment:** It essentially represents the spread of depolarization through the AV node, the bundle of His, and bundle branches and normally is displayed as a flat line.
- **QRS complex:** This represents the activity of the ventricles and seen as three closely related waves on the ECG (Q, R and S wave). Normally, the duration which is measured from the beginning of the QRS complex to its end is 0.08 to 0.10 seconds.
- **ST segment:** It starts at the end of the S wave and finishes at the start of the T wave. It is an isoelectric line that represents the time between depolarization and repolarization of the ventricles.
- **T wave:** It is seen as a small wave after the QRS complex that represents ventricular repolarization. Normal duration of the T wave can vary close to the 160 ms.
- **QT interval:** It starts at the beginning of the QRS complex and finishes at the end of the T wave. It represents the time taken for the ventricles to depolarize and then repolarize.
- **U wave:** Normally hypothesized to be caused by repolarization of interventricular septum.

4.2.2 The Lead System

ECG is recorded by a device with electrodes attaching to the body skin. The whole set-up is generally called the lead system. The first 'standard' ECG consisted of only the three limb leads is introduced by Einthoven in 1931. Around 3 years later, Wilson and Goldberger developed the ECG lead system. The standard 12-lead combined the lead systems from Einthoven, Wilson and Goldberger Figure 4.3 and Figure 4.5 illustrate the standard 12-lead system on a human body. Commonly, 10 electrodes attached to the body are used to form 12 ECG leads, with each lead measuring a specific electrical potential difference. Each lead refers to an imaginary line between two different ECG electrodes. All leads can be divided into three main types which are limb leads, augmented limb leads and precordial leads.

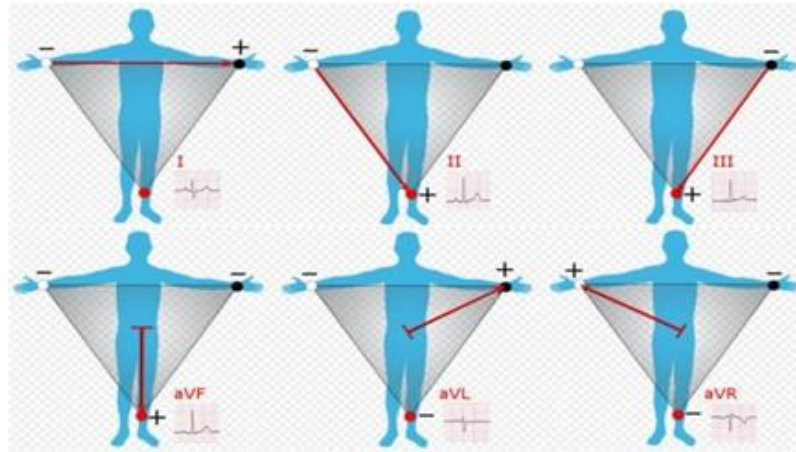


Figure 4.3: The limb lead diagram(Top) and the augmented lead diagram(Bottom)

Leads I, II and III are called the limb leads. The top three of Figure 4.3 illustrates these limb leads on a human body. The electrodes that form these signals are located on the limbs – one on each arm and one on the left leg. Lead I is the voltage between the (positive) left arm (LA) electrode and right arm (RA) electrode. Lead II is calculated as the voltage between left leg (LL) electrode and right arm (RA) electrode and lead III represents the voltage from left leg (LL) electrode to left arm (LA) electrode.

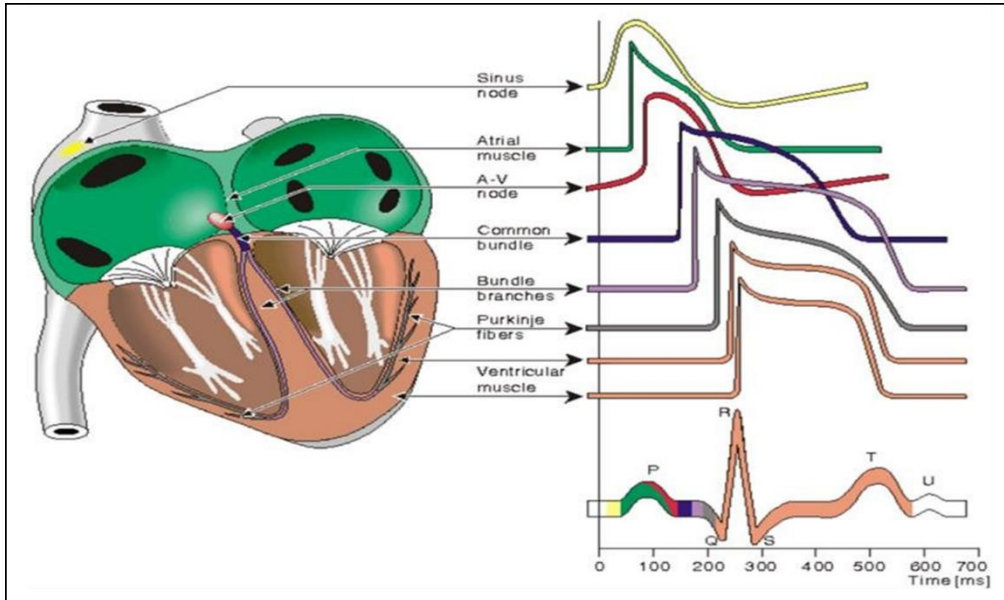


Figure 4.4: The electrical conduction system of the heart

The heart deals with detailed morphology and timing of ECG signal from different parts of the heart and ultimately leading to the form of an ECG heartbeat. The mathematical equations of the voltage recorded for three limb leads are shown respectively.

4.3 Cardiovascular Disease

4.3.1 Premature (Extra) Beats and Arrhythmia

Premature Beats consist of Premature (Extra) Atrial Contractions (PACs) and Premature Ventricular Contractions also called Ventricular Premature beats (VPBs). PACs are extra beats that occur in the atria. In most cases, they are harmless without any symptoms. Therefore normally they do not require treatment. On the other hand, premature beats that originate in the lower chambers (ventricles) are defined as PVCs. The main characteristic of PVCs is the compensatory pause following the contraction as shown below in figure 4.5. The extra action potential causes the SA node to become refractory to generating its next scheduled beat, and thus it must “skip a beat” and will resume exactly two P-P intervals after the last normal sinus beat.

For some people, PVCs can happen because of too much exercise and they are also relative with stress and too much caffeine or nicotine. Sometimes, heart disease or electrolyte imbalance can also lead to PVCs. However, in most people, the hazard of PVCs can be negligible.

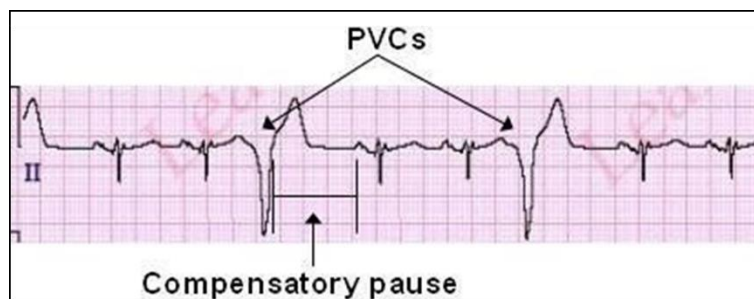


Figure 4.5: The ECG record of PVCs from Lead II

As mentioned earlier, a healthy heart supplies the human body with the right amount of blood at the rate needed to work normal. In general, a heartbeat should be 60 to 100 times per minute. However, if disease or injury weakens the human heart, the rhythm of heart will be abnormal, which is known as arrhythmia. There are two major categories of arrhythmia based on the heart rate. Tachycardia mentioned in [13] is one broad category of arrhythmia that is a heart rate that exceeds the normal resting rate. On the other hand, arrhythmia which leads to the slow heart rate is defined as bradycardia. In the following subsection, we present discussion about different types of arrhythmias

4.3.2 Supraventricular Arrhythmias

Another category of arrhythmias called Supraventricular Arrhythmias can be divided into Atrial Fibrillation , Atrial Flutter and Paroxysmal Supraventricular Tachycardia (PSVT) as in [22]. AF is a common type of arrhythmias that involves a fast and irregular heart rhythm and can cause the abnormal contraction of atria.

4.3.3 Atrial Fibrillation (AF)

In AF, the heart's electrical impulses originate in another position of the atria instead of the sinoatrial node. Moreover, the heart's electrical impulses travel through the whole atria rapidly rather than a normal way, which may cause the atrial wall beats very fast as shown in ECG of figure 4.6. In result, the atria are not capable of pumping blood into the ventricle in a right way. The major causes of AF are high blood pressure, inflammation and coronary heart disease. In addition, an early research proposed that the risk of AF is strongly associated with age. The advancing age can lead to the higher incidence of AF.

4.3.4 Atrial Flutter (AFL)

AFL is an arrhythmia that is like the AF. AF usually involves a rapid and irregular heart rhythm, however, in AFL, the electrical impulses travel throughout the atria with fast and regular heart rhythm. It occurs due to one or more rapid circuits in the atrium and usually can be found in people with heart disease or after cardiac surgery [12] can be seen in below figure 4.6.

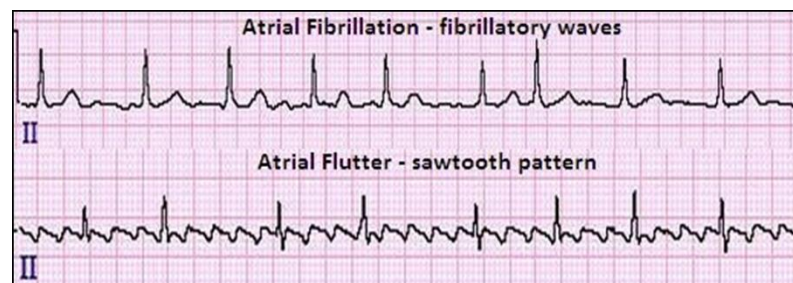


Figure 4.6: The ECG records of Atrial Fibrillation and Atrial Flutter from Lead II

4.3.5 Ventricular Arrhythmias

Ventricular Arrhythmias are the most dangerous category of arrhythmias and normally composed of Ventricular Tachycardia (VT) and Ventricular Fibrillation (VF). VT[15] can occur with many

variations of the QRS morphology, depending on where the arrhythmia originates, which sometimes makes diagnosis on ECG challenging.

4.3.6 Myocardial infarction

Myocardial infarction (MI), commonly known as a heart attack, is a critical condition where blood flow to the heart muscle is obstructed, causing tissue damage. An ECG (electrocardiogram) is a vital tool in diagnosing and managing MI by capturing the heart's electrical activity.

During an MI, specific changes appear on an ECG. The most characteristic feature is the elevation of the ST segment, known as ST-elevation myocardial infarction (STEMI). This elevation indicates a complete blockage of a coronary artery. Additionally, the development of new Q waves suggests necrosis of heart tissue. In contrast, non-ST-elevation myocardial infarction (NSTEMI) might show ST depression or T wave inversion, indicating a partial blockage.

The ECG records these anomalies across different leads, helping pinpoint the infarct's location. For instance, ST elevation in leads II, III, and aVF suggests an inferior MI, typically involving the right coronary artery. Conversely, changes in leads V1-V4 indicate an anterior MI, affecting the left anterior descending artery.

Prompt ECG analysis enables rapid intervention, such as thrombolytic therapy or percutaneous coronary intervention, restoring blood flow and minimizing heart damage. Continuous ECG monitoring during and after treatment is crucial for detecting arrhythmias, another common complication of MI.

4.3.7 Patient with History of Myocardial Infraction

A patient with a history of myocardial infarction (MI) requires comprehensive and ongoing medical management to prevent further cardiac events and maintain optimal heart health. Post-MI care involves a multifaceted approach encompassing lifestyle modifications, medication adherence, and regular monitoring.

Lifestyle changes are pivotal in secondary prevention. Patients are advised to adopt a heart-healthy diet rich in fruits, vegetables, whole grains, and lean proteins while minimizing saturated fats, salt, and sugars. Regular physical activity, tailored to the patient's capabilities, aids in maintaining cardiovascular fitness and managing weight. Smoking cessation is crucial, as tobacco use significantly increases the risk of recurrent MI.

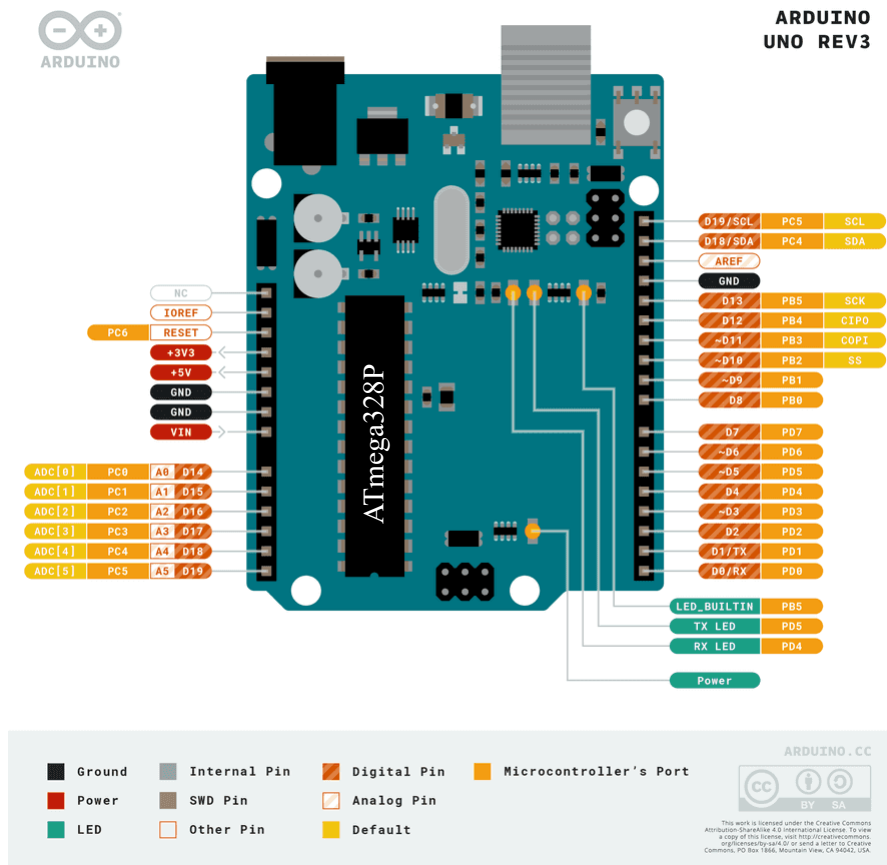
Medication regimens typically include antiplatelet agents like aspirin, beta-blockers, statins, and ACE inhibitors or ARBs. These medications help prevent blood clots, reduce blood pressure, lower cholesterol levels, and improve heart function. Adherence to these medications is essential to minimize the risk of subsequent cardiac events.

Regular follow-up appointments with healthcare providers are necessary to monitor the patient's condition and adjust treatment plans as needed. Periodic assessments, including ECGs, stress tests, and echocardiograms, help evaluate heart function and detect any early signs of complications.

Emotional and psychological support is also important, as many patients experience anxiety or depression after MI. Cardiac rehabilitation programs offer a structured environment for physical activity, education, and counselling, supporting both physical and mental recovery.

4.4 Hardware and software

4.4.1 Arduino Uno



Arduino UNO is a microcontroller board based on the ATmega328P. It has 14 digital input/output pins (of which 6 can be used as PWM outputs), 6 analog inputs, a 16 MHz ceramic resonator, a USB connection, a power jack, an ICSP header and a reset button. It contains everything needed to support the microcontroller; simply connect it to a computer with a USB cable or power it with a AC-to-DC adapter or battery to get started. The Arduino UNO board is designed in a way that allows it to be reset by software running on a connected computer. One of the hardware flow control lines (DTR) of the ATmega8U2/16U2 is connected to the reset line of the ATmega328 via a 100 nano-farad capacitor. When this line is asserted (taken low), the reset line drops long enough to reset the chip. The Arduino Software (IDE) uses this capability to allow you to upload code by simply pressing the upload button in the interface toolbar. This means that the bootloader can have a shorter timeout, as the lowering of DTR can be well-coordinated with the start of the upload.

4.4.2 AD8232 ECG Module

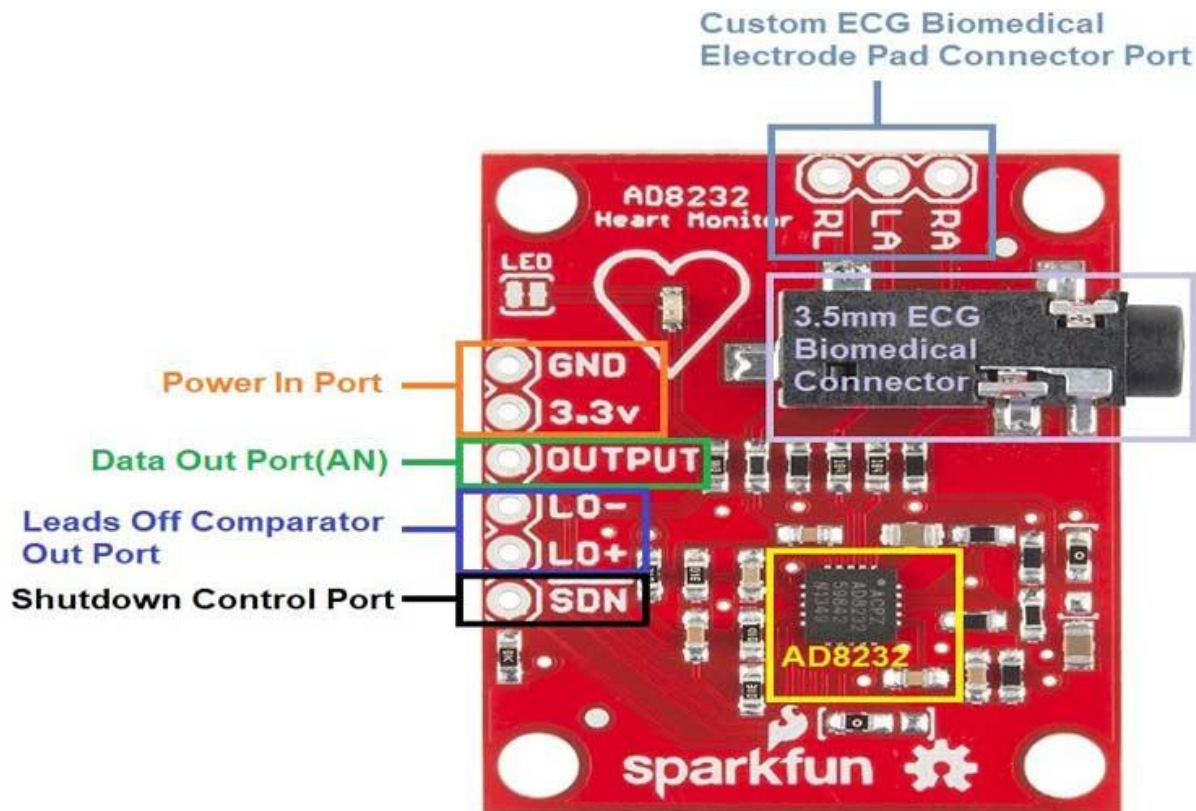


Figure 4.7(b) AD8232 ECG Module Pinout Diagram

AD8232 is the ECG sensor module with an integrated circuit. It is useful for measuring and monitoring the electrical activity of the heart or heartbeat. It is made up of a high-quality chip. It can be useful for

building fitness, heart exercises for heart rate monitoring. It can be used as a portable ECG or remote heart rate monitoring. In addition, it is used to help diagnose various heart conditions.

This sensor is a cost-effective board. This electrical activity can be charted as an ECG or Electrocardiogram and output as an analog reading. However, ECGs can be extremely noisy, and the AD8232 Single Lead Heart Rate Monitor acts as an op-amp to help obtain a clear signal from the PR and QT Intervals easily.

The AD8232 is an integrated signal conditioning block for ECG and other bio-potential measurement applications. It is designed to extract, amplify, and filter small bio-potential signals in the presence of noisy conditions, such as those created by motion or remote electrode placement.

4.4.3 HC-05 Bluetooth Module

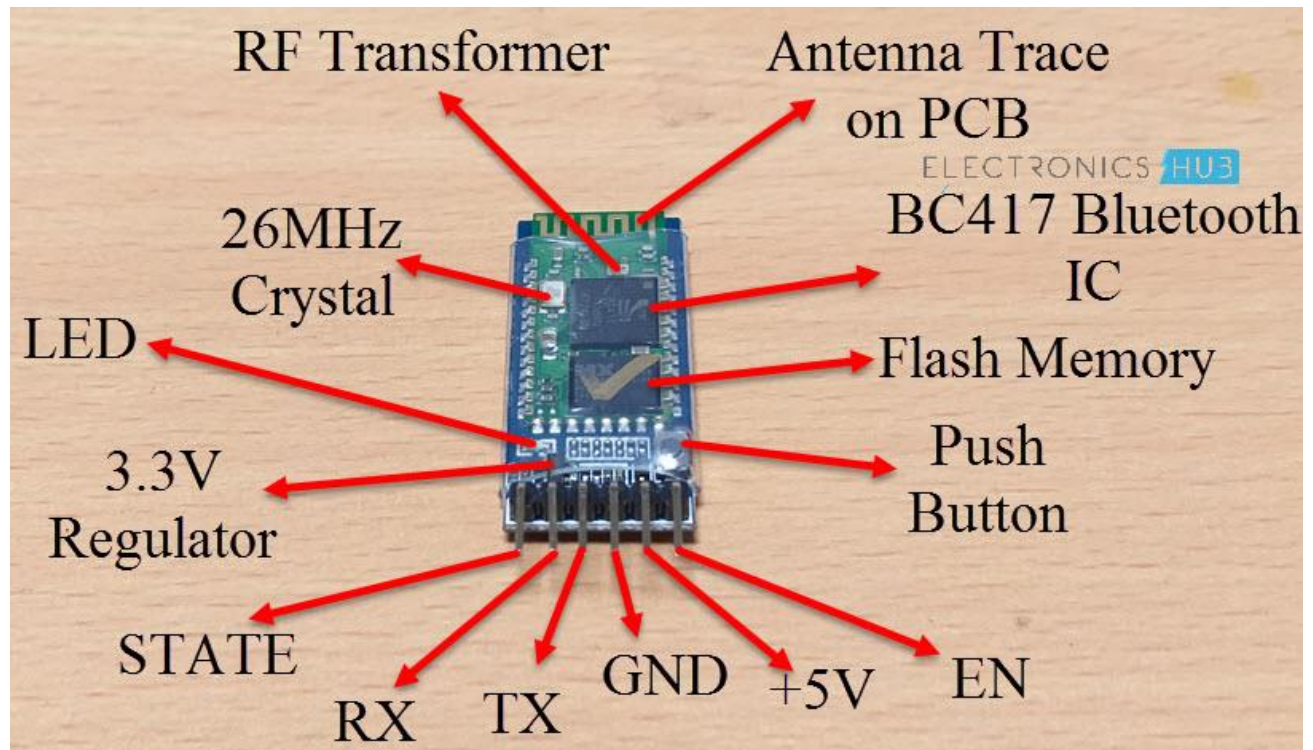


Figure 4.7(c) Bluetooth Module

This module is based on BC417 Single Chip Bluetooth IC that is compliant with Bluetooth v2.0 standard and with support for both UART and USB interfaces.

HC-05 Bluetooth Module, to be precise, comes with the BC417 IC along with a flash memory. Such Modules come as surface mount board and several third-party manufacturers use these board to build a more complete system with necessary pins and components.

The HC-05 Module supports for UART, USB as well as SPI communication and depending on the application, necessary pins can be used. In my case, the board uses the UART communication.

Four pins are sufficient for successfully enabling a wireless communication link but the modules produced now-a-days come with six pins namely: VCC, GND, TX, RX, EN and STATE. An important point to remember is the HC-05 Bluetooth Module works on a logic level of 3.3V. Hence, a 3.3V Regulator is used on the board.

NOTE: The button present on the board is used to configure the Bluetooth Module in AT Command Mode. This part of the Module is not discussed in this project.

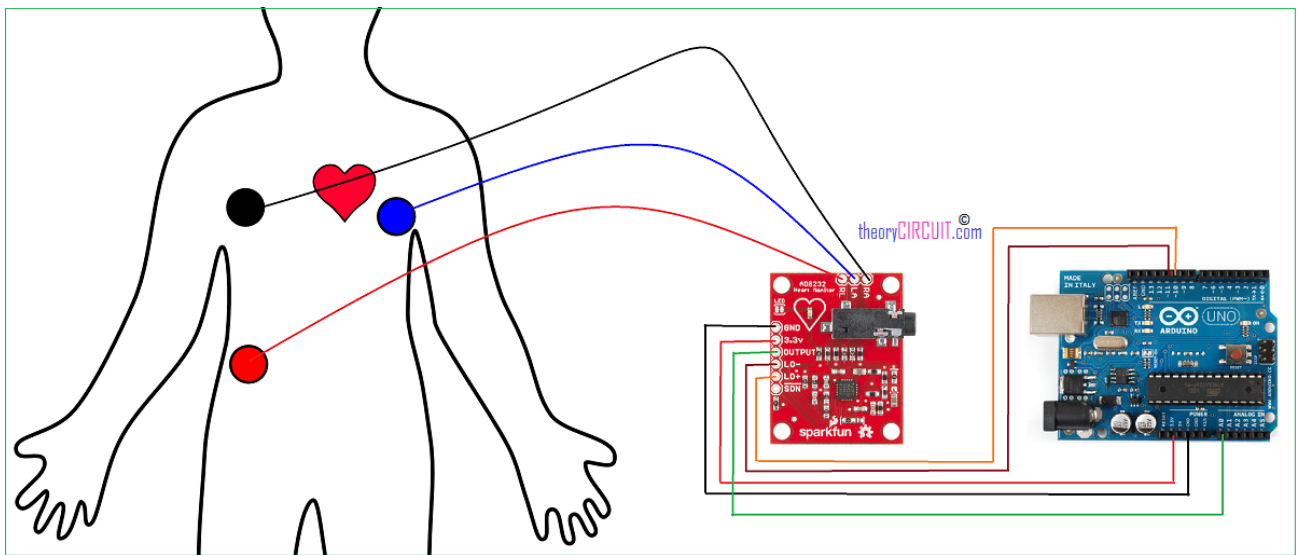


Figure 4.7(d) AD8232 ECG Sensor & Arduino Interfacing with ECG

The figure 4.7 illustrates the setup for measuring heart rate using an Arduino. Electrodes are placed on the chest and limbs of a person to detect electrical signals from the heart. These signals are transmitted to a heart rate sensor module (AD8232), which processes the data. The sensor module is connected to an Arduino Uno board, which captures and digitizes the signals for further analysis or display. The wiring diagram shows the connections between the electrodes, the heart rate sensor, and the Arduino, demonstrating a straightforward method for creating a basic ECG monitor using easily accessible electronic components.

4.5 Signal Processing

4.5.1 ECG Preprocessing

Pre-processing is clearly required in ECG applications since ECG is affected by various kind of noises. Normally, these noises can be divided into four main types which are power line interference, Electromyography (EMG) artefact, electrode contact noise and baseline wandering [10]. The following sections will give introduction of these noises and the corresponding removal methods.

4.5.2 ECG Feature Extraction

The most useful information in the ECG is normally derived from the amplitudes and intervals that are defined by the fiducial points of ECG wave (e.g. onset, offset, peak). In general, these features are used to classify the normal and abnormal heartbeats (Arrhythmias) and accordingly produce an alarm. Therefore, it is necessary to find some methods to extract such as ECG features. Recently, Many research papers have presented various method achieve feature extraction of ECG [5]. In general, there are two main groups of ECG feature extraction algorithms, which are QRS detection algorithms and other wave delineation algorithms.

4.5.3 QRS detection algorithm

The QRS detection is the primary task of the ECG feature extraction because it is the most characteristic waveform of the ECG signal. The properties of the QRS complex, which are high amplitude and specific morphology, makes QRS detection easier than the other waves. Thus, it is usually used as a reference in the ECG signal. Pan and Tompkins first introduced a conventional method named pan-tompkins algorithm on the digital band-pass filter to detect QRS complex .There are other attempts for QRS detection based on time domain morphology and gradient method. In addition, the wavelet transform is widely used to detect the QRS complex. The study suggested using the discrete wavelet transform to identify the QRS complex. Recently, many research presented QRS detection algorithms with lower computation burden. A QRS detection method with low- complexity in individual ECG beat named hybrid feature extraction algorithm that based on the discrete wavelet transform was introduced in Mazomenos et al. Other WT-based algorithms defined a real-time QRS detection method based on phase portraits and box-scoring calculation. Optimized adaptive thresholding , iterative state machines, and moving-average filters are also presented to detect QRS complex.

4.5.4 Wave delineation algorithm

The main objective of these algorithms is to determine the peaks and boundaries of the individual QRS complexes, P and T waves. In general, this algorithm in figure 4.8 estimate the location of the QRS complex initially and then define the time windows before and after the QRS fiducial point (onset and offset) to detect the P and T waves. The locations of P and T waves onsets, peaks and offsets are difficult to determine because the amplitudes of the noises can be higher than the P and T waves. In this situation, it is necessary to apply some techniques to enhance the characteristic features of the P and T waves. Some methods are used to extract ECG wave features in the frequency band, such as discrete wavelet transform (DWT).

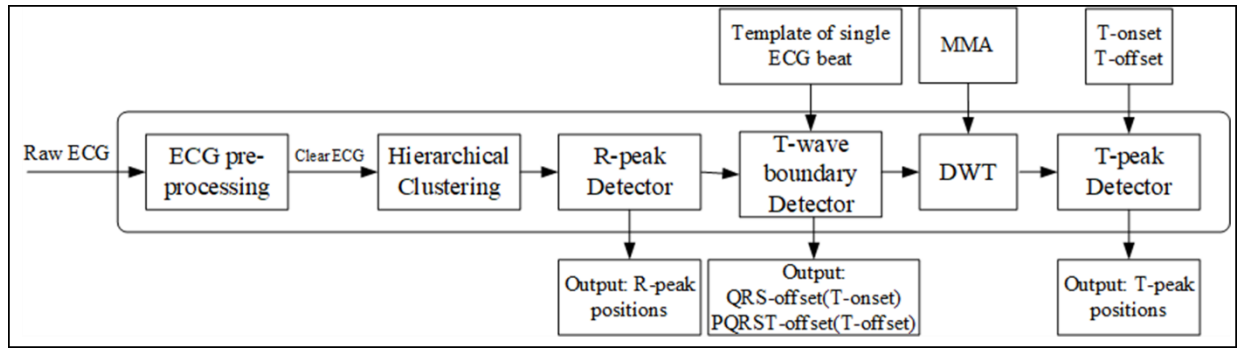


Figure 4.8: The system overview of the proposed algorithm

QRS Complex Detection: Use QRS complex detection algorithms to identify the QRS waveform in the ECG signal. These algorithms typically involve thresholding, peak detection, or template matching techniques to locate the R peaks, which mark the onset and offset of the QRS complex. In the context of predicting heart status, the T wave is one of the important components analysed in an ECG. Abnormalities or variations in the T wave can provide valuable information about potential cardiac conditions or disturbances.

4.6 Conclusion

In this chapter, the basic anatomy of the heart, its electrical condition system, as well as ECG and lead system have been introduced. Signal processing methods for ECG pre- processing and feature extraction have also been described. We have broadly reviewed a range of arrhythmia detection, classification and predication algorithms pertinent to the thesis and highlighted their shortcomings.

A new algorithm based on the hierarchical clustering and discrete wavelet transform for the automated delineation of the ECG fiducial points (R and T peaks). Our algorithm has been validated on the MIT-BIH arrhythmia database, QT database and UHS database. The results show that the performance of algorithm is generally better than other referenced algorithms. The use of hierarchical clustering allows for identifying the R clusters and determining the R peaks with high accuracy of 99.83% based on MIT-BIH database. The combination of DWT and MMA analysis detect the T peaks with high sensitivity of 99.91% based on QT database. Also our algorithm can effectively reduce the TWOS and the number of R-peak false positive detection. However, since the template of each ECG record in MIT- BIH and QT databases was randomly chosen from healthy parts, the performance of our algorithm might be changed with different templates. Therefore, the robustness of our algorithm should be further evaluated with the different healthy templates.

CHAPTER 5

A Deep Neural Network Approach: Convolutional Neural Network

5.1 Introduction

Deep neural networks (DNNs) are a type of artificial neural network (ANN) that are capable of learning complex representations of data. They are called "deep" because they typically have many layers (i.e., are deep) compared to traditional neural networks, which may have only one or two layers. DNNs have been highly successful in a variety of applications, including image recognition, speech recognition, natural language processing, and even games like Go and chess.

The basic building block of a DNN is a neuron, which receives input from other neurons and applies a mathematical function to produce an output. The outputs of neurons in one layer become inputs to the neurons in the next layer, and this process continues until the final output is produced. Each layer of neurons learns to recognize and represent different features of the input data, with the earlier layers learning simpler features and the later layers learning more complex ones.

DNNs are trained using a process called backpropagation [21], which involves adjusting the weights of the connections between neurons to minimize a loss function that measures the difference between the network's output and the desired output. This process is typically performed using large amounts of labelled data, with the network adjusting its weights based on the errors it makes on the training data.

One of the key advantages of DNNs is their ability to automatically learn useful features from raw data, without the need for manual feature engineering. This has led to breakthroughs in areas such as image and speech recognition, where DNNs have achieved levels of accuracy that were previously thought impossible.

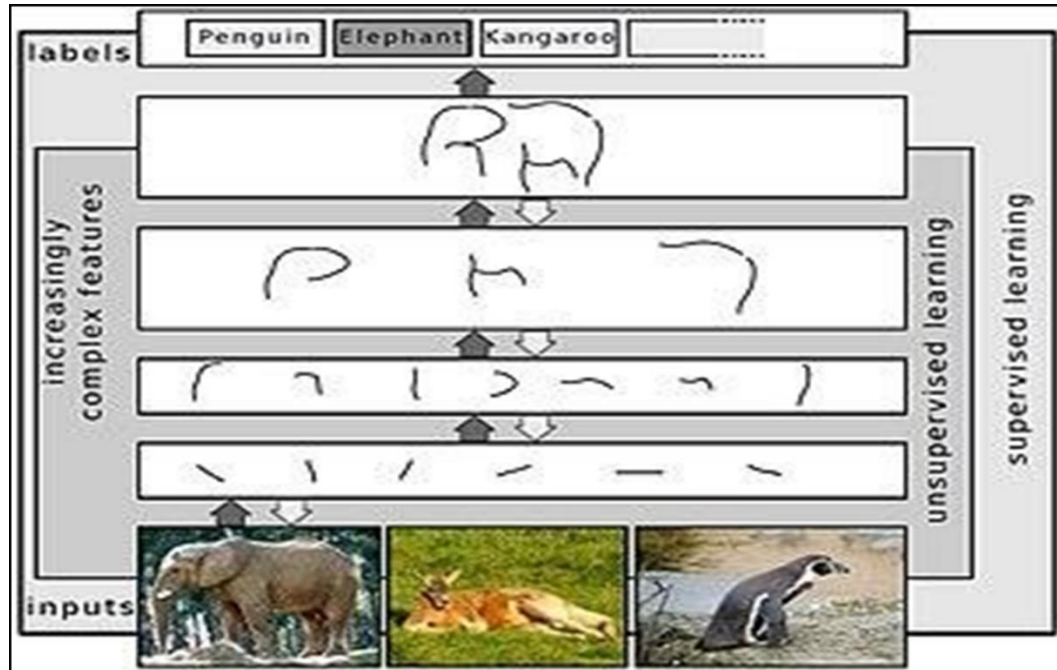


Figure 5.1: Researching images on multiple layers of deep network

The main purpose of a neural network is to receive a set of inputs, perform progressively complex calculations on them, and give output to solve real world problems like classification. We restrict ourselves to feed forward neural networks.

We have an input, an output, and a flow of sequential data in a deep network. One example of DL is the mapping of a photo to the name of the person(s) in photo as they do on social networks and describing a picture with a phrase is another recent application of DL.

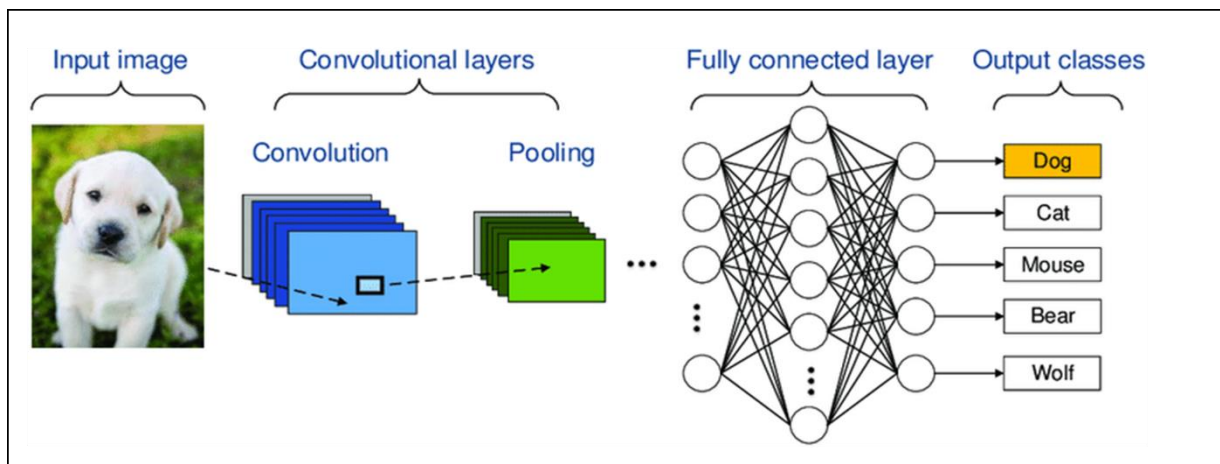


Figure 5.2: Example implementation of deep network

5.2 Classification Of Deep Neural Networks

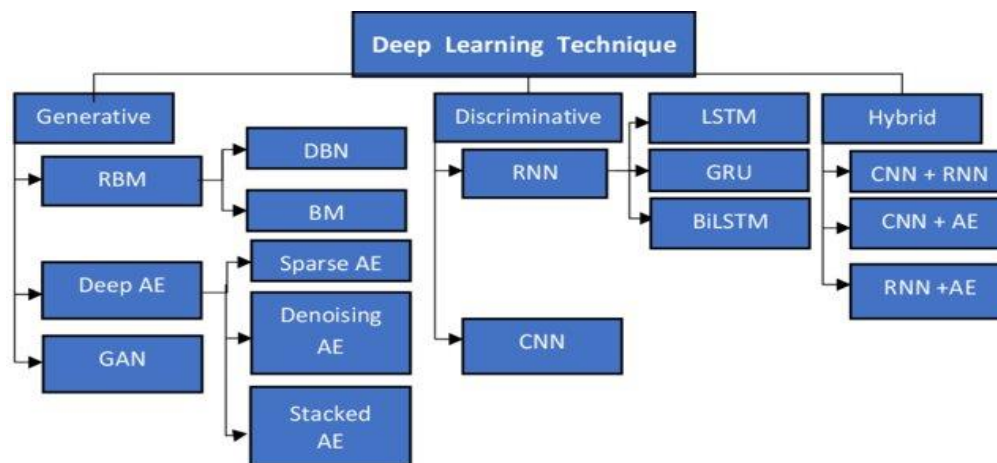


Figure 5.3 : Categories Of Deep Neural Network

Three following types of deep neural networks are popularly used today:

1. Multi-Layer Perceptrons (MLP)
2. Convolutional Neural Networks (CNN)
3. Recurrent Neural Networks (RNN)

Multilayer Perceptrons (MLPs): A multilayer perceptrons a class of a feedforward artificial neural network. MLPs models are the most basic deep neural network, which is composed of a series of fully connected layers. Today, MLP machine learning methods can be used to overcome the requirement of high computing power required by modern deep learning architectures. Each new layer is a set of nonlinear functions of a weighted sum of all outputs (fully connected) from the prior one.

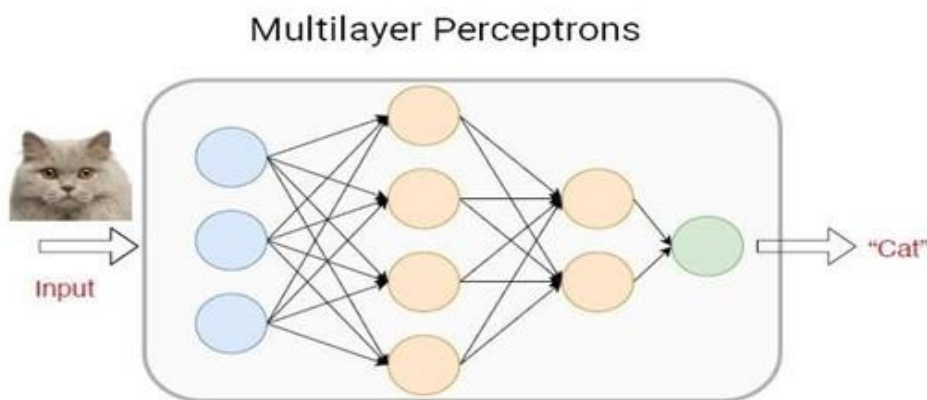


Figure 5.4: Multilayer Perceptrons

Recurrent Neural Network (RNN): A recurrent neural network is another class of artificial neural networks that use sequential data feeding. RNNs have been developed to address the time-series problem of sequential input data. The input of RNN consists of the current input and the previous samples. Therefore, the connections between nodes form a directed graph along a temporal sequence. Furthermore, each neuron in an RNN owns an internal memory that keeps the information of the computation from the previous samples.

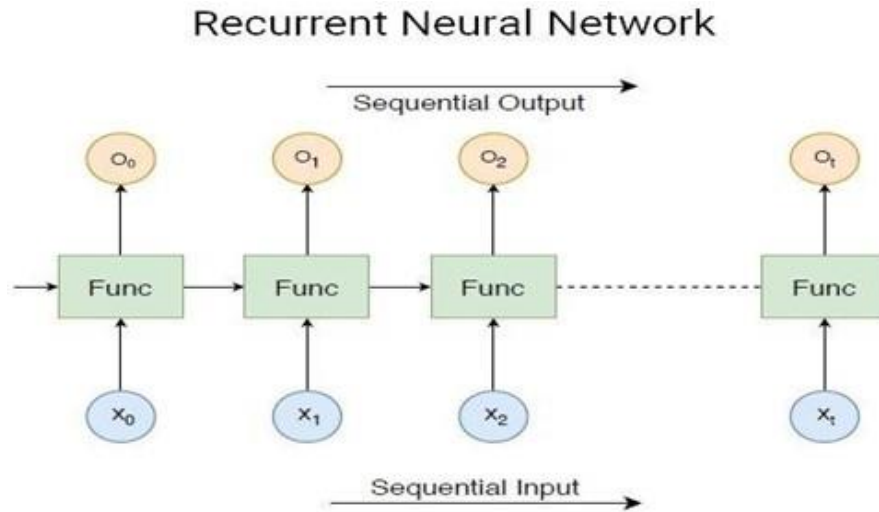


Figure 5.5: Working of RNN

5.3 Convolutional Neural Network (CNN)

Convolutional neural network is another class of deep neural networks. CNNs are most commonly employed in computer vision. Given a series of images or videos from the real world, with the utilization of CNN, the AI system learns to automatically extract the features of these inputs to complete a specific task, e.g., image classification, face authentication, and image semantic segmentation. Different from fully connected layers in MLPs, in CNN models, one or multiple convolution layers extract the simple features from input by executing convolution operations. Each layer is a set of nonlinear functions of weighted sums at different coordinates of spatially nearby subsets of outputs from the prior layer, which allows the weights to be reused.

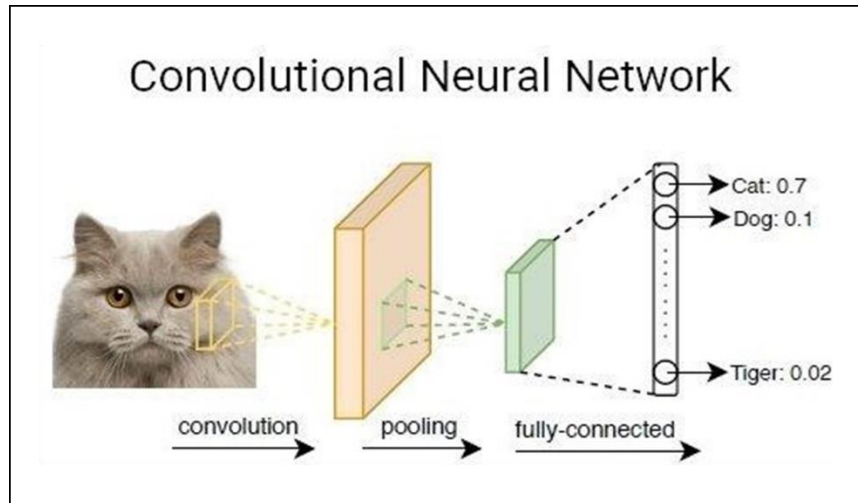


Figure 5.6: Steps of Convolutional Neural Networks

5.3.1 Working Of CNN

Convolutional neural networks are distinguished from other neural networks by their superior performance with image, speech, or audio signal inputs. They have three main types of layers, which are:

- Convolutional layer
- Pooling layer
- Fully-connected (FC) layer

The convolutional layer is the first layer of a convolutional network. While convolutional layers can be followed by additional convolutional layers or pooling layers, the fully-connected layer is the final layer. With each layer, the CNN increases in its complexity, identifying greater portions of the image. Earlier layers focus on simple features, such as colours and edges. As the image data progresses through the layers of the CNN, it starts to recognize larger elements or shapes of the object until it finally identifies the intended object.

1. Convolutional Layer

The convolutional layer is the core building block of a CNN, and it is where the majority of computation occurs. It requires a few components, which are input data, a filter, and a feature map. Let's assume that the input will be a colour image, which is made up of a matrix of pixels in 3D. This means that the input will have three dimensions—a height, width, and depth—which correspond to

RGB in an image. We also have a feature detector, also known as a kernel or a filter, which will move across the receptive fields of the image, checking if the feature is present. This process is known as a convolution.

The feature detector is a two-dimensional (2-D) array of weights, which represents part of the image. While they can vary in size, the filter size is typically a 3x3 matrix; this also determines the size of the receptive field. The filter is then applied to an area of the image, and a dot product is calculated between the input pixels and the filter. This dot product is then fed into an output array. Afterwards, the filter shifts by a stride, repeating the process until the kernel has swept across the entire image. The final output from the series of dot products from the input and the filter is known as a feature map, activation map, or a convolved feature.

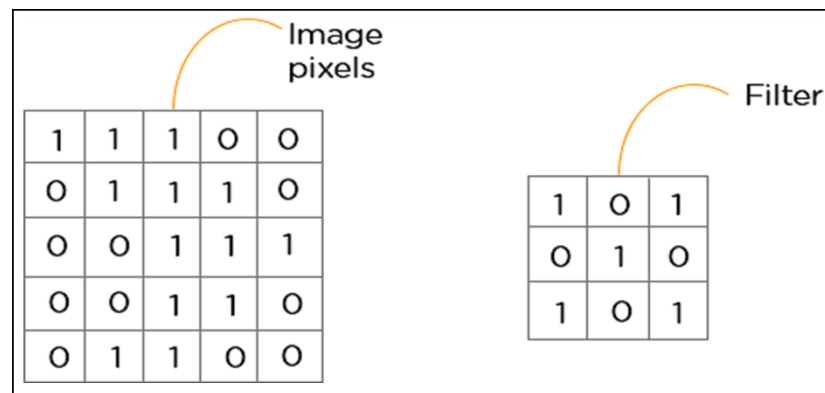


Figure 5.7: View of Convolutional layer

After each convolution operation, a CNN applies a Rectified Linear Unit (ReLU) transformation to the feature map, introducing nonlinearity to the model. As we mentioned earlier, another convolution layer can follow the initial convolution layer. When this happens, the structure of the CNN can become hierarchical as the later layers can see the pixels within the receptive fields of prior layers.

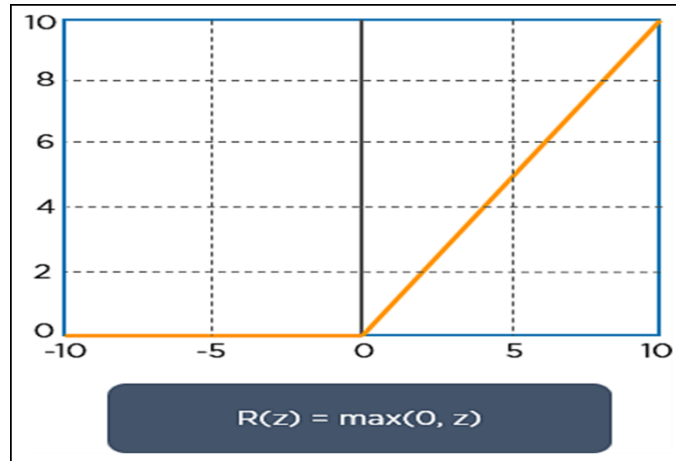


Figure 5.8: Graphical demonstration of ReLu layer

As an example, let's assume that we're trying to determine if an image contains a bicycle. You can think of the bicycle as a sum of parts. It is comprised of a frame, handlebars, wheels, pedals, etc. Each individual part of the bicycle makes up a lower-level pattern in the neural net, and the combination of its parts represents a higher-level pattern, creating a feature hierarchy within the CNN.

2. Pooling Layer

Pooling layers, also known as down sampling, conducts dimensionality reduction, reducing the number of parameters in the input. Similar to the convolutional layer, the pooling operation sweeps a filter across the entire input, but the difference is that this filter does not have any weights. Instead, the kernel applies an aggregation function to the values within the receptive field, populating the output array. There are two main types of pooling:

Max pooling: As the filter moves across the input, it selects the pixel with the maximum value to send to the output array. As an aside, this approach tends to be used more often compared to average pooling [19].

Average pooling: As the filter moves across the input, it calculates the average value within the receptive field to send to the output array.

While a lot of information is lost in the pooling layer, it also has a number of benefits to the CNN. They help to reduce complexity, improve efficiency, and limit risk of overfitting.

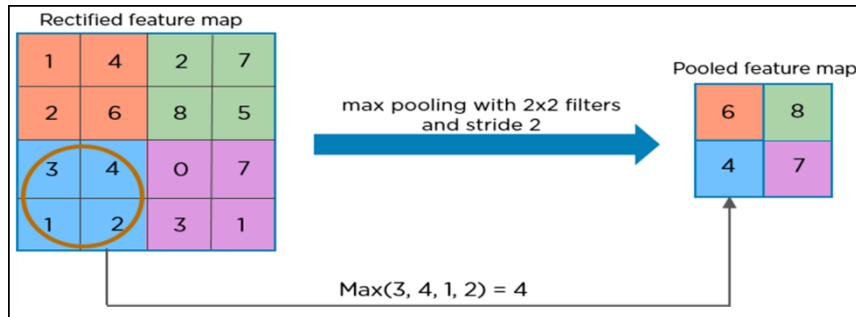


Figure5.9 Demonstration of pooling layer

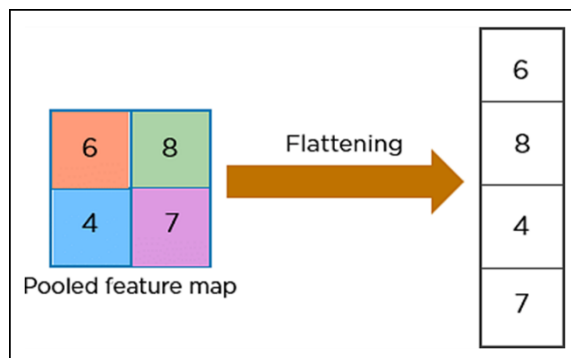


Figure 5.10: Flattening

Flattening refers to the process of reshaping the multi-dimensional output of the convolutional and pooling layers into a one-dimensional vector. This operation is typically performed before feeding the data into fully connected layers for classification or further processing.

3.Fully-Connected Layer

The name of the full-connected layer aptly describes itself. As mentioned earlier, the pixel values of the input image are not directly connected to the output layer in partially connected layers. However, in the fully-connected layer, each node in the output layer connects directly to a node in the previous layer.

This layer performs the task of classification based on the features extracted through the previous layers and their different filters. While convolutional and pooling layers tend to use ReLu functions, FC layers usually leverage a SoftMax activation function to classify inputs appropriately, producing a probability from 0 to 1.

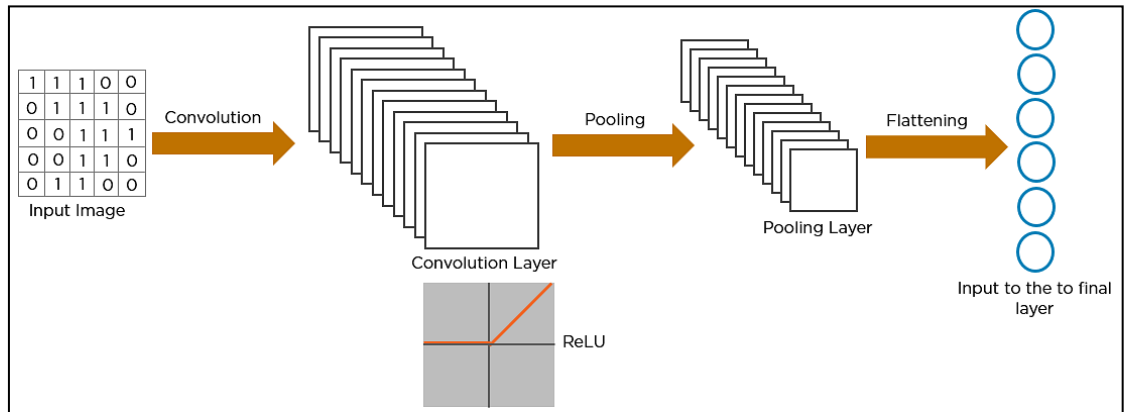


Figure 5.11: Architecture of CNN and step by step approach

CNN example implementation of a picture of bird

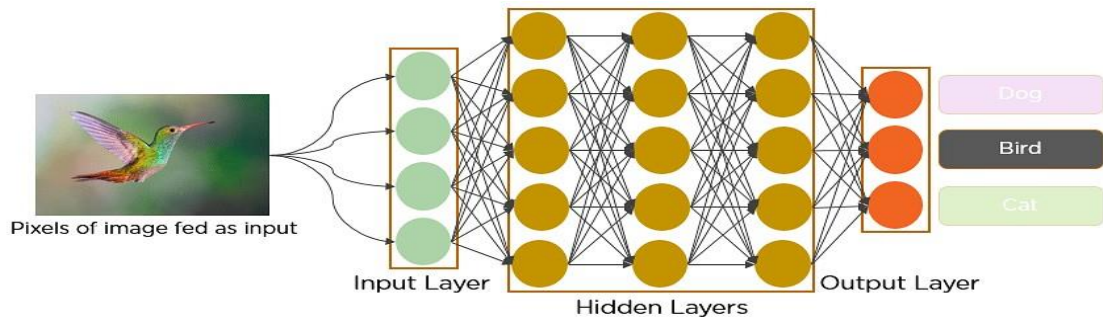


Figure: 5.12 (a) Bird image as input

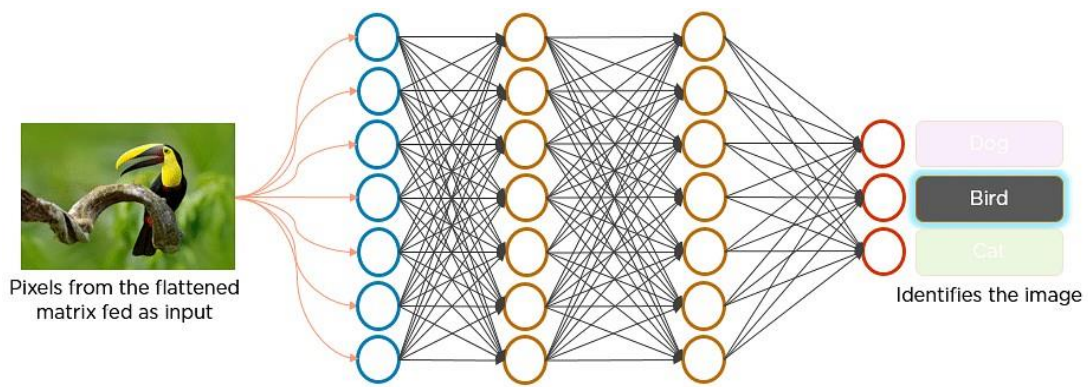


Figure: 5.12 (b): Implementation of CNN on example

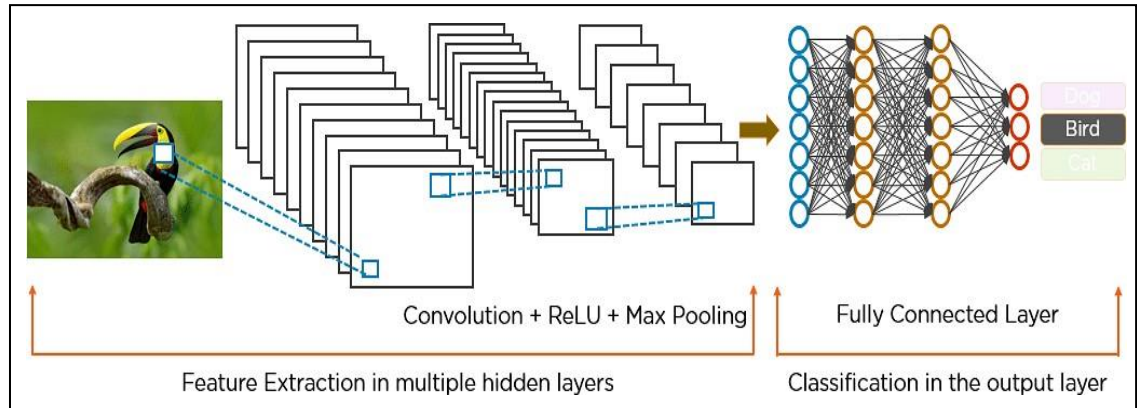


Figure: 5.12(c) : Bird as output of example

Here's how exactly CNN recognizes a bird:

- I. The pixels from the image are fed to the convolutional layer that performs the convolution operation
- II. It results in a convolved map
- III. The convolved map is applied to a ReLU function to generate a rectified feature map
- IV. The image is processed with multiple convolutions and ReLU layers for locating the features
- V. Different pooling layers with various filters are used to identify specific parts of the image.
- VI. The pooled feature map is flattened and fed to a fully connected layer to get the final output.

5.3.2 Calculations

Table 5.1: CNN Calculations

Convolution	$z^l = h^{l-1} * W^l$
Max Pooling	$h^l_{xy} = \max_{i=0..s, j=0..s} h^{l-1}(x+i)(y+j)$
Fully-connected layer	$z_l = W_l * h_{l-1}$
ReLu(Rectifier)	$\text{ReLU}(z_i) = \max(0, z_i)$
Softmax	$\text{softmax}(z_i) = e^{z_i} / \sum_j e^{z_j}$

5.4 Conclusions

Chapter 5 concludes, deep neural networks (DNNs) and convolutional neural networks (CNNs) are two powerful types of artificial neural networks that have revolutionized the field of artificial intelligence. DNNs are capable of learning complex representations of data, while CNNs are specifically designed for processing images and other types of multidimensional data.

Both DNNs and CNNs have been highly successful in a variety of applications, including image recognition, speech recognition, natural language processing, and even games like Go and chess. They have enabled breakthroughs in areas such as healthcare, finance, and transportation, and have the potential to transform many other fields as well. One of the key advantages of DNNs and CNNs is their ability to automatically learn useful features from raw data, without the need for manual feature engineering. This has made them highly effective at handling large volumes of complex data, and has enabled them to achieve levels of accuracy that were previously thought impossible.

However, there are also challenges associated with DNNs and CNNs [20], such as the need for large amounts of labelled data, the potential for overfitting, and the difficulty of interpreting the learned representations. There are also ethical considerations, such as the potential for bias and unfairness in decision-making based on these models. Overall, DNNs and CNNs are powerful tools for building complex models that can learn to recognize and classify complex patterns in data. With ongoing research and development, they have the potential to transform many areas of science and industry, and open up new opportunities for artificial intelligence.

CHAPTER 6

Methods and Materials

6.1 Introduction

This Chapter introduces the reader to the dataset. The entire MIT-BIH Arrhythmia Database is now freely available in PhysioBank. This introduction describes how the database records were obtained, and discusses the characteristics of the recorded signals. In most records, the upper signal is a modified limb lead II (MLII), obtained by placing the electrodes on the chest. The lower signal is usually a modified lead V1 as for the upper signal, the electrodes are also placed on the chest. This configuration is routinely used by the BIH Arrhythmia Laboratory. Normal QRS complexes are usually prominent in the upper signal. The lead axis for the lower signal may be nearly orthogonal to the mean cardiac electrical axis, however (i.e., normal beats are usually biphasic and may be nearly isoelectric). Thus normal beats are frequently difficult to discern in the lower signal, although ectopic beats will often be more prominent for. A notable exception is record 114, for which the signals were reversed. Since this happens occasionally in clinical practice, arrhythmia detectors should be equipped to deal with this situation. In records 102 and 104 [9], it was not possible to use modified lead II because of surgical dressings on the patients; modified lead V5 was used for the upper signal in these records.

6.1.1 Data Description

The MIT-BIH Arrhythmia Database contains 48 half-hour excerpts of two-channel ambulatory ECG recordings, obtained from 47 subjects studied by the BIH Arrhythmia Laboratory between 1975 and 1979. Twenty-three recordings were chosen at random from a set of 4000 24-hour ambulatory ECG recordings collected from a mixed population of inpatients (about 60%) and outpatients (about 40%) at Boston's Beth Israel Hospital; the remaining 25 recordings were selected from the same set to include less common but clinically significant arrhythmias that would not be well-represented in a small random sample.

The recordings were digitized at 360 samples per second per channel with 11-bit resolution over a 10 mV range. Two or more cardiologists independently annotated each record; disagreements were resolved to obtain the computer-readable reference annotations for each beat (approximately 110,000 annotations in all) included with the database.

This directory contains the entire MIT-BIH Arrhythmia Database. About half (25 of 48 complete records, and reference annotation files for all 48 records) of this database has been freely available here since PhysioNet's inception in September 1999. The 23 remaining signal files, which had been available only on the MIT-BIH Arrhythmia Database CD-ROM, were posted here in February 2005.

6.1.2 Files

Computerized electrocardiogram interpretation plays a critical role in the clinical ECG workflow¹. Widely available digital ECG data and the algorithmic paradigm of deep learning² present an opportunity to substantially improve the accuracy and scalability of automated ECG analysis. However, a comprehensive evaluation of an end-to-end deep learning approach for ECG analysis across a wide variety of diagnostic classes has not been previously reported. Developed a deep neural network to classify 12 rhythm classes using 91,232 single-lead ECGs from 53,549 patients who used a single-lead ambulatory ECG monitoring device. When validated against an independent test dataset annotated by a consensus committee of board-certified practicing cardiologists, the DNN achieved an average area under the receiver operating characteristic curve (ROC) of 0.97. The average F1 score, which is the harmonic mean of the positive predictive value and sensitivity, for the DNN (0.837) exceeded that of average cardiologists (0.780). With specificity fixed at the average specificity achieved by cardiologists, the sensitivity of the DNN exceeded the average cardiologist sensitivity for all rhythm classes. These findings demonstrate that an end-to-end deep learning approach can classify a broad range of distinct arrhythmias from single-lead ECGs with high diagnostic performance similar to that of cardiologists. If confirmed in clinical settings, this approach could reduce the rate of misdiagnosed computerized ECG interpretations and improve the efficiency of expert human ECG interpretation by accurately triaging or prioritizing the most urgent conditions.

6.1.3 Model

In this project, a model which can diagnose irregular heart rhythms, also known as arrhythmias, from single-lead ECG signals better than a cardiologist is presented.

The model is mainly made of 3 blocks, the input plus a reset block, and a reset loop-block and a output block. One of the 1-D convolution layer in the loop-block reduces the half of the size of the layer per bi-loops. Thus, during 15 loops, it shrink down the input size of the signal to 1/256 times. And its filter length gets double up with zero pad every 4 loops. Then the filter grows from 1 to 32 at the input block and enlarges from 32 to 256 at the loop- block.

We train a 34-layer convolutional neural network (CNN) to detect arrhythmias in arbitrary length ECG time-series.

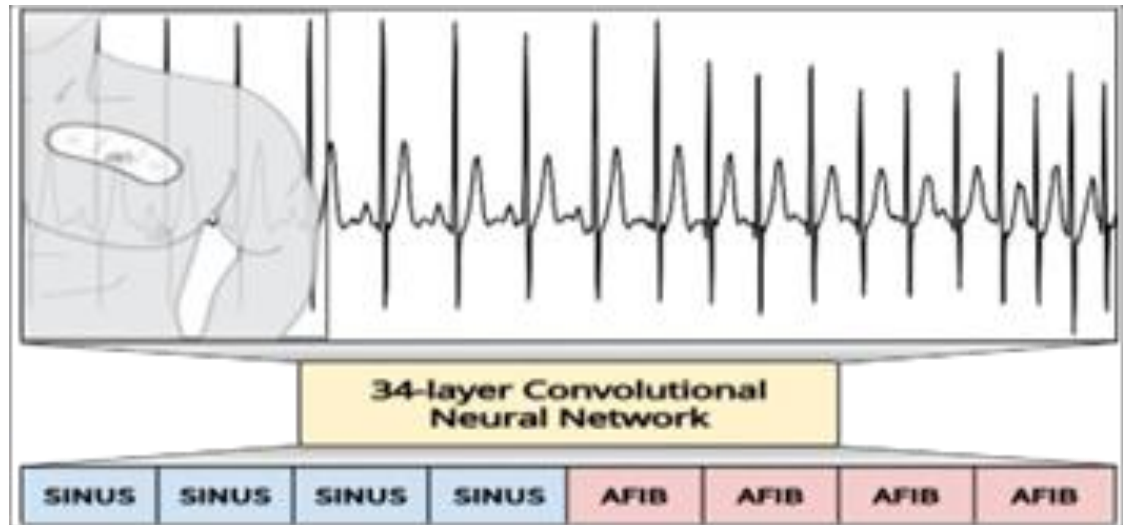


Figure: 6.1 (a) : 34-layer Convolutional Neural Network-2

The network takes as input a time-series of raw ECG signal, and outputs a sequence of label predictions. The 30 second long ECG signal is sampled at 200Hz, and the model outputs a new prediction once every second. We arrive at an architecture which is 33 layers of convolution followed by a fully connected layer and a SoftMax. To make the optimization of such a deep model tractable, we use residual connections and batch-normalization. The depth increases both the non-linearity of the computation as well as the size of the context window for each classification decision.

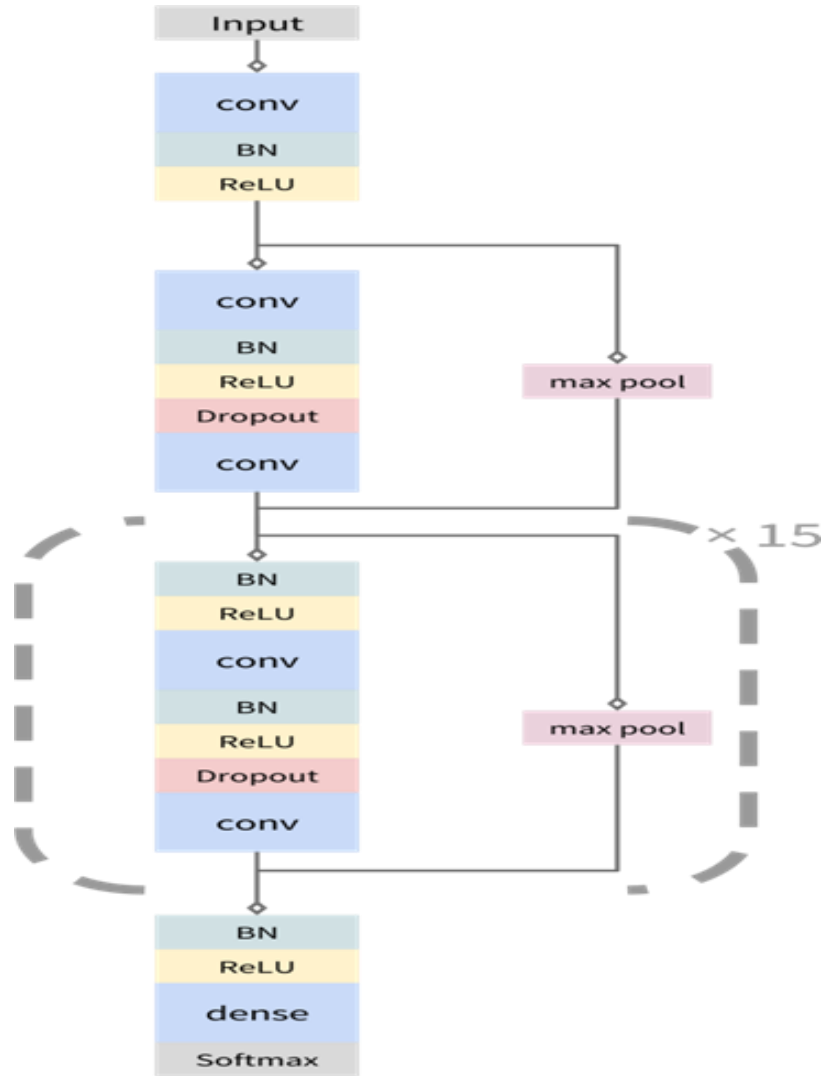


Figure: 6.1 (b) : 34-layer Convolutional Neural Network-2

Collected and annotate a dataset of 64,121 ECG records from 29,163 patients.

The ECG data is sampled at a frequency of 200 Hz and is collected from a single-lead, noninvasive and continuous monitoring device called the Zio Patch which has a wear period up to 14 days. Each ECG record in the training set is 30 seconds long and can contain more than one rhythm type. Each record is annotated by a clinical ECG expert: the expert highlights segments of the signal and marks it as corresponding to one of the 14 rhythm classes.

We collect a test set of 336 records from 328 unique patients. For the test set, ground truth annotations for each record were obtained by a committee of three board-certified cardiologists; there are three committees responsible for different splits of the test set. The cardiologists discussed each individual record as a group and came to a consensus labeling. For each record in the test set we also collect 6 individual annotations from cardiologists not participating in the group. This is used to assess performance of the model compared to an individual cardiologist.

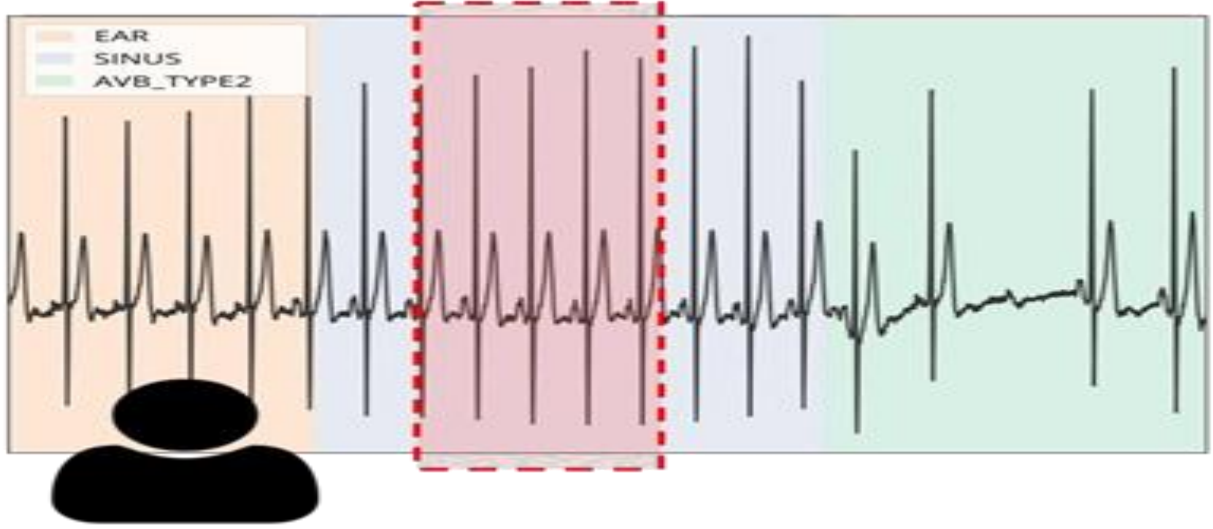


Figure: 6.2 :Zoomed View Of ECG

The model outperforms the cardiologist average on both the Sequence and Set F1 metrics. The Sequence F1 measures the average overlap between the prediction and the ground truth sequence labels, while the Set F1 measures the average overlap between the unique class labels from the prediction and those from the ground truth.

6.1.4 Limitations

One of the most common confusions is between Ectopic Atrial Rhythm (EAR) and the sinus rhythm (SINUS). The main distinguishing criteria for this rhythm is an irregular P wave. This can be subtle to detect especially when the P wave has a small amplitude or when noise is present in the signal. Similarly, Supraventricular Tachycardia and Atrial Fibrillation are often confused with Atrial Flutter which is understandable given that they are all atrial arrhythmias. Idioventricular Rhythm is sometimes mistaken as Ventricular Tachycardia (VT), which again makes sense given that the two only differ in heart-rate and are difficult to distinguish close to the 100 beats per minute delineation.

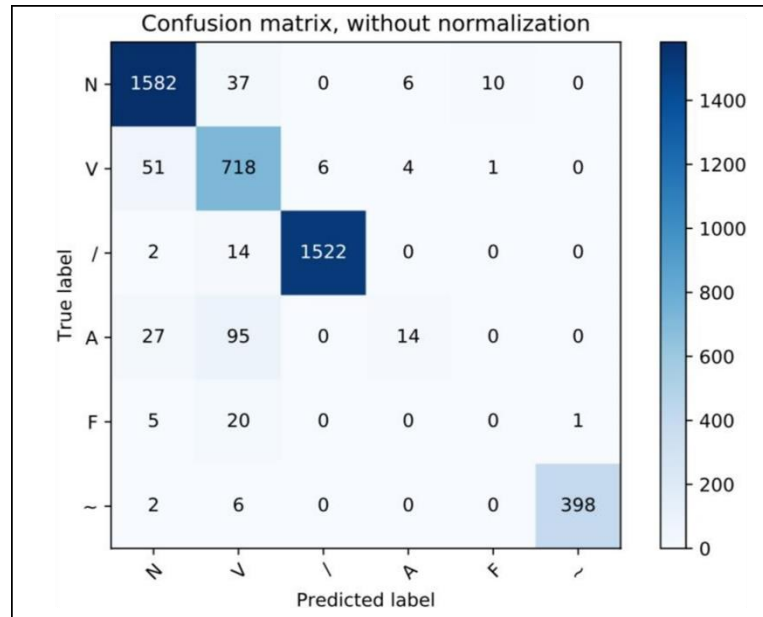


Figure: 6.3 : Graph between true label and predicted label

6.2 Understand literatures and the result-analysis

Deep learning and classifications.

The pattern recognition using deep convolutional neural network is indisputably good. It shows in various complicated image recognitions or even sound recognition. It is obvious it is going to be so good at least as the similar level of human being.

What matters is if we have enough data, and how we can preprocess the data properly for machine to learn effectively. Collecting serviceable ECG data is not easy . What we obtained is only the data provided by Physionet and irhythm tech.

From the below table, as compared with hand written digit recognition data, CINC data are 5 times smaller in total array size and other datasets are even smaller than CINC data. Besides, the most of the ECG data are normal data, so the abnormal data to train the ill heart signal are seriously lack to perform well.

Table 6.1 : ECG database

	number	signal size	total size	sampling rate	total time
CINC	8000	9000	72000000	300	240000
MIT-BIH	47	650000	30550000	360	84861
IRHYTHM	328	3000	984000	200	4920
MNIST	60000	784	47040000	None	None
Stanford	80000	3000	240000000	200	1200000

The highest precision score in CINC 2017 competition is about 83 percent, and the F1 score or other micro metric would be even worse since the number 83 must have been helped by normal signals which are major of the data.

6.3 Cleaning data

Compare to about 8 thousands CINC data, MIT-BIH contains only 47 signals, however, each signal is long, with 650000 samples for each signal. Besides, the signals have an annotation of samples for each peak. In other words, we have many labels for each signal. Thus, I collect out all the high peaks of the heart beat signals and matched each one of the peaks to a sliced 256-sized sample about a peak. Then, this 256-sized sample is well-fit into the model without flattening layer.²

Ironically, a lot of N-labeled data cause a trouble for this model. The normal sinus rhythmic data is easy to gain, so most of collected data are N-labeled. In the trainer, to reduce the cost(loss) of the model, finding the weights of the model to make N to be more precise is way more efficient than finding weights of the model to be balanced classifier. Overall, they have better recall and F1 scores in general labels.

These five kinds of heart beat signals are secured to have more data than others for a qualified classifier. I have also removed L, R labels. These are branch bundle block beats which were originally labeled as normal beats before.

There are 5 types, MLII, V1, V2, V4, V5 in MIT-BIH dataset. I mainly trained and tested MLII and V1 features among them because other features do not have many data. The signal would be recorded differently in magnitude and shapes by devices and the where to measure. V1 has a potential to classify branch bundle block beats discussed above. I do not know much about it and leave room for you think about.

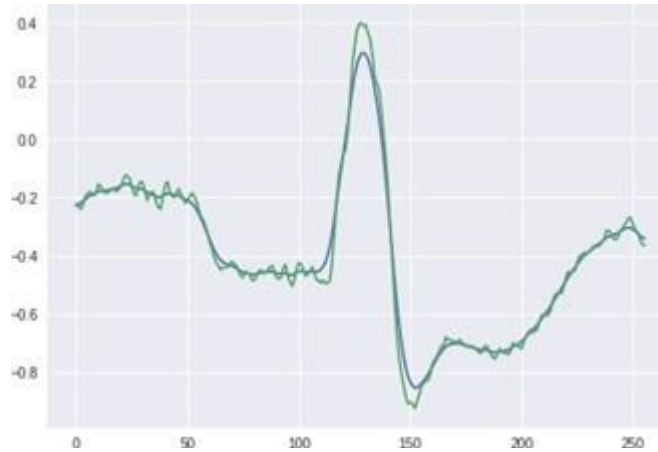


Figure 6.4: ECG Waveform -QRS peak

6.4 Libraries Used

When working on heart status prediction with ECG signals using CNNs, several libraries can be useful for implementing and training the models. Here are some commonly used libraries:

- 1. TensorFlow:** TensorFlow is a popular open-source deep learning library that provides comprehensive support for building and training neural networks, including CNNs. It offers high level APIs, such as Keras, for creating and training CNN models efficiently.
- 2. PyTorch:** PyTorch is another widely used deep learning library that offers dynamic computational graphs and a flexible framework for building CNN models. It provides extensive support for GPU acceleration and includes modules for creating CNN architectures, optimizing models, and managing data.
- 3. Keras:** Keras is a high-level deep learning library that runs on top of TensorFlow or other backends such as PyTorch or Theano. It offers a user-friendly API for designing and training CNN models with ECG signals. Keras provides easy-to-use abstractions for building CNN architectures, managing layers, and optimizing models.
- 4. Scikit-learn:** Scikit-learn is a popular machine learning library in Python that includes various tools for data preprocessing, model evaluation, and performance metrics. It provides functionalities for data preprocessing (e.g., scaling, feature selection), model evaluation (e.g., cross-validation), and performance metrics (e.g., accuracy, precision, recall) that are useful in heart status prediction tasks.

5. NumPy: NumPy is a fundamental library for numerical computing in Python. It provides support for large, multi-dimensional arrays and mathematical operations required for data manipulation and processing in ECG signal analysis. NumPy is often used in conjunction with other libraries for data preprocessing and manipulation tasks.

6. Pandas: Pandas is a powerful library for data manipulation and analysis in Python. It offers data structures, such as Data Frames, that simplify handling and processing tabular data, including ECG signal data. Pandas can be useful for loading, cleaning, and organizing ECG datasets before training CNN models.

7. Matplotlib: These libraries provide plotting and visualization capabilities in Python. They can be used to visualize ECG signals, model training progress, and performance evaluation results. Matplotlib provides a flexible and comprehensive set of plotting functions, while Seaborn offers high-level, aesthetically pleasing statistical visualization.

8. Flask: Flask is a micro web framework that is designed to be lightweight and modular, which means that you can add only the features that you need. It is built on top of the Werkzeug WSGI toolkit and the Jinja2 template engine. Flask is very flexible and can be used for a wide range of web applications, from simple static websites to complex web applications with dynamic content and user authentication.

6.5 Conclusion

Chapter 6 concludes that MIT-BIH Arrhythmia Database offers a comprehensive collection of ECG recordings critical for advancing arrhythmia detection technologies. The data includes modified limb lead II and lead V1 signals, collected in a way that closely mimics clinical practices. Challenges arise from records with reversed signals or surgical dressings, requiring adaptive arrhythmia detectors. The database, with its high sampling rate and thorough annotations by cardiologists, serves as a robust foundation for developing and testing deep learning models. The introduction of a 34-layer convolutional neural network (CNN) for arrhythmia classification demonstrates significant advancements. This model, validated against cardiologist-annotated datasets, achieves superior performance metrics, showcasing deep learning's potential to match or exceed human expertise. Limitations include confusions between specific arrhythmias due to subtle signal differences, emphasizing the need for more comprehensive datasets and enhanced preprocessing techniques. The integration of libraries like TensorFlow, PyTorch, and Keras, alongside tools like Scikit-learn and Pandas, facilitates effective model development and data handling. Visualization tools and web frameworks further support the deployment and analysis of ECG data. Overall, while the current model shows promise, future improvements hinge on increasing data diversity, refining hyperparameters, and enhancing the robustness of signal interpretation, potentially revolutionizing automated ECG analysis and cardiac care.

CHAPTER 7

RESULTS AND CONCLUSIONS

7.1 Introduction

This chapter describes about the diagnosis and results. Here, the final deployment is discussed.

7.2 Predict Model

	precision	recall	f1-score	support
0	0.95	0.97	0.96	1635
1	0.81	0.92	0.86	780
2	1.00	0.99	0.99	1538
3	0.58	0.10	0.18	136
4	0.00	0.00	0.00	26
5	1.00	0.98	0.99	406
micro avg	0.94	0.94	0.94	4521
macro avg	0.72	0.66	0.66	4521
weighted avg	0.93	0.94	0.93	4521

Confusion matrix, without normalization						
[[1582	37	0	6	10	0]	
[51	718	6	4	1	0]	
[2	14	1522	0	0	0]	
[27	95	0	14	0	0]	
[5	20	0	0	0	1]	
[2	6	0	0	0	398]]	
F1 score:	[0.95762712	0.85988024	0.99282453	0.175	0.	0.98881988]
AUC for N:	0.9999909698392632					
AUC for V:	0.9999880960883747					
AUC for /:	1.0					
AUC for A:	1.0					
AUC for ~:	1.0					

Figure 7.1: Overall View Of Predictions

The trained model, can predict CINC2017 competition trainset and compare the result with its true labels.

In the figure 7.1, it shows part of predictions of slices (it was too long to be all captured) and there are many labels for abnormal beats. However, the most predicted label is still “N” with 55.6 percentages, and the second predicted label is matched to the true label, “A”, with 25.9 percentages. The third probability is 8.2 percentage and much smaller than the second

```

The 78/103-record classified as A with 99.0% certainty
The 80/103-record classified as A with 98.3% certainty
The 81/103-record classified as Noise with 99.1% certainty
The 83/103-record classified as Ventricular with 73.0% certainty
The 84/103-record classified as A with 99.1% certainty
The 86/103-record classified as Noise with 99.4% certainty
The 91/103-record classified as A with 39.5% certainty
The 93/103-record classified as A with 73.5% certainty
The 94/103-record classified as A with 98.2% certainty
The 95/103-record classified as A with 93.9% certainty
The 97/103-record classified as Noise with 81.6% certainty
The average of the predict is: [[5.5603498e-01 8.2084917e-02 3.8246583e-04 2.5941592e-01 9.7061656e-03
 9.2375450e-02]]
The most predicted label is N with 55.6% certainty
The second predicted label is A with 25.9% certainty
The original label of the record is A

```

Figure 7.2: Result Of Training the data

7.3 Flask web app

Flask and Streamlit are popular frameworks for developing web applications in Python. Flask is a micro-framework that provides the essential tools to build web apps, offering flexibility and control over components like routing and templates. It's lightweight and highly customizable, making it suitable for a wide range of applications. It is used in this project figure 7.3 for real time processing of ECG Signal.

Streamlit, on the other hand, is designed specifically for data scientists. It allows for rapid creation of data-driven web applications with minimal code. Streamlit's simplicity in integrating visualizations and interactive widgets makes it ideal for prototyping and sharing data analysis projects. Both frameworks streamline web app development but serve distinct purposes and audiences. It is used in this for ECG processing from dataset given images.

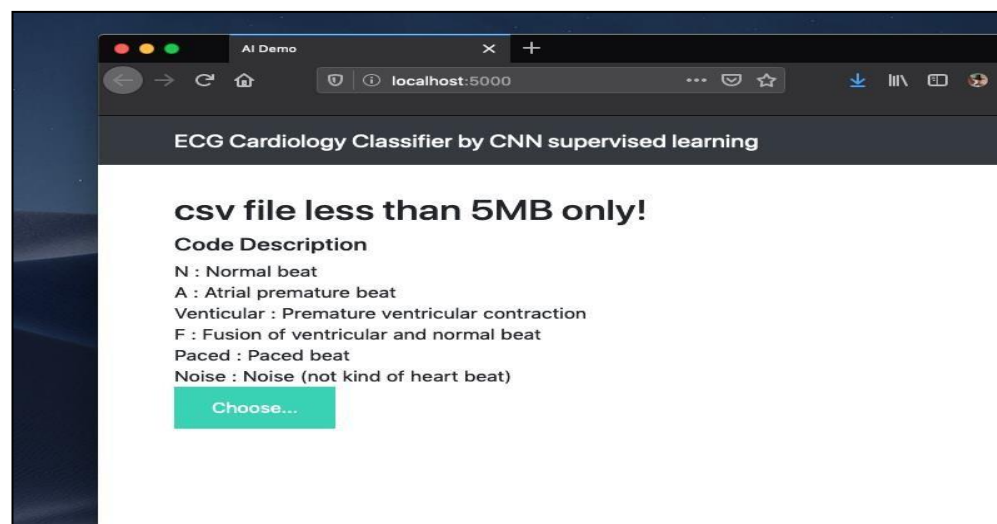


Figure 7.3: Flask Web View result of deployment

A1						
	A	B	C	D	E	F
1	360	sampling rate				
2	995					
3	995					
4	995					
5	995					
6	995					
7	995					
8	995					
9	1000					
10	997					
11	995					
12	994					
13	992					
14	993					
15	992					
16	989					

figure 7.4 Sampled data in CSV format.

In the figure 7.5, the signals have 57 peaks, and the signal of each peak was estimated by the trained model. Then the whole signals consist of 27 normal beats, 6 ventricular beats, 2 atrial beats, and 22 noises signals. If the signals represent noise, the most of the signals should have been estimated as noise. Then, the patient with the signals must be healthy and have a good ventricular contraction.

ECG Cardiology Classifier by CNN supervised learning

Help

csv file less than 5MB only!

Code Description

N : Normal beat
A : Atrial premature beat
Ventricular : Premature ventricular contraction
F : Fusion of ventricular and normal beat
Paced : Paced beat
Noise : Noise (not kind of heart beat)

Choose...

Result: 57 parts of the divided data were estimated as the followings with paired probabilities. (Noise:96.0%)(N:100.0%)(Noise:84.8%)(N:99.9%)(Noise:82.2%)(N:62.6%)(Noise:83.0%)(N:99.4%)(Ventricular:64.0%)(N:96.5%)(N:98.3%)(Noise:88.3%)(N:84.9%)(Noise:82.8%)(N:98.8%)(Ventricular:51.9%)(N:100.0%)(Noise:83.7%)(N:100.0%)(Noise:69.6%)(N:99.1%)(Ventricular:58.8%)(N:73.9%)(Noise:77.7%)(N:72.3%)(Noise:71.5%)(N:98.2%)(Noise:82.5%)(N:92.8%)(Noise:95.5%)(N:97.9%)(Noise:75.9%)(N:83.5%)(Noise:84.0%)(N:100.0%)(Noise:65.4%)(N:94.4%)(Noise:62.3%)(A:62.7%)(Noise:82.2%)(A:69.7%)(Noise:61.4%)(N:67.2%)(Noise:82.4%)(N:65.4%)(Ventricular:65.6%)(N:98.0%)(Noise:60.2%)(N:95.7%)(Noise:77.2%)(N:95.8%)(Ventricular:56.4%)(N:98.3%)(Ventricular:69.4%)(N:98.8%)(Noise:83.1%)(N:98.1%)27-N, 6-Ventricular, 0-Paced, 2-A, 0-F, 22-Noise

Figure 7.5: Output Result

7.4 Heartbeat Classification

Normal Heartbeat Classification:

Through meticulous analysis of electrocardiographic data, characterized the normal heartbeat, delineating its key features and electrical patterns with precision. By achieving a comprehensive understanding of normal sinus rhythm, a foundational framework for distinguishing cardiac abnormalities and advancing diagnostic accuracy.

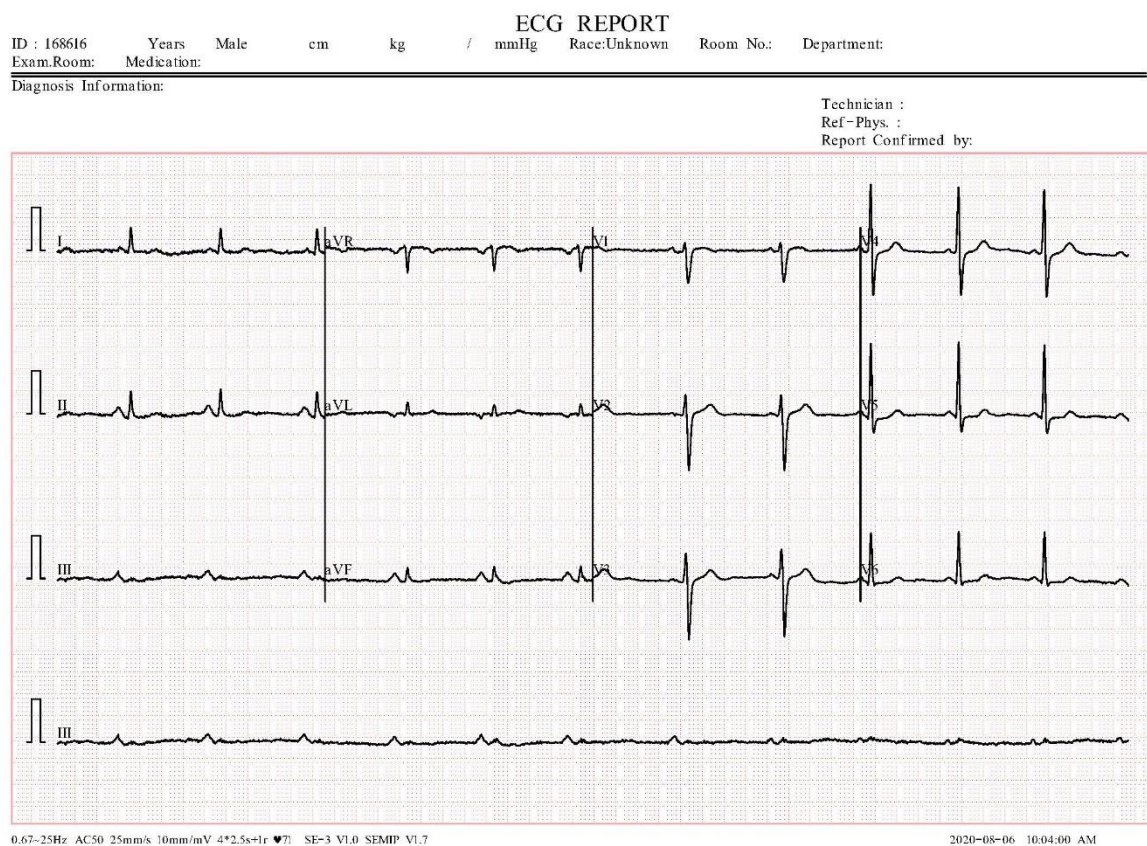


Figure 7.6: Uploaded ECG waveform of normal heartbeat input

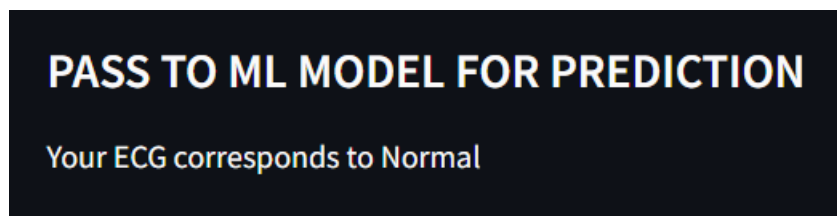


Figure 7.7: Received Output of the waveform

Abnormal heartbeat classification:

By scrutinizing various arrhythmic patterns and their clinical implications, discerned the intricate nuances of abnormal heart rhythms. Through this exploration, enhanced ability to recognize and interpret irregularities in cardiac electrical activity, thereby bolstering diagnostic acumen and facilitating targeted interventions for patients with arrhythmias.

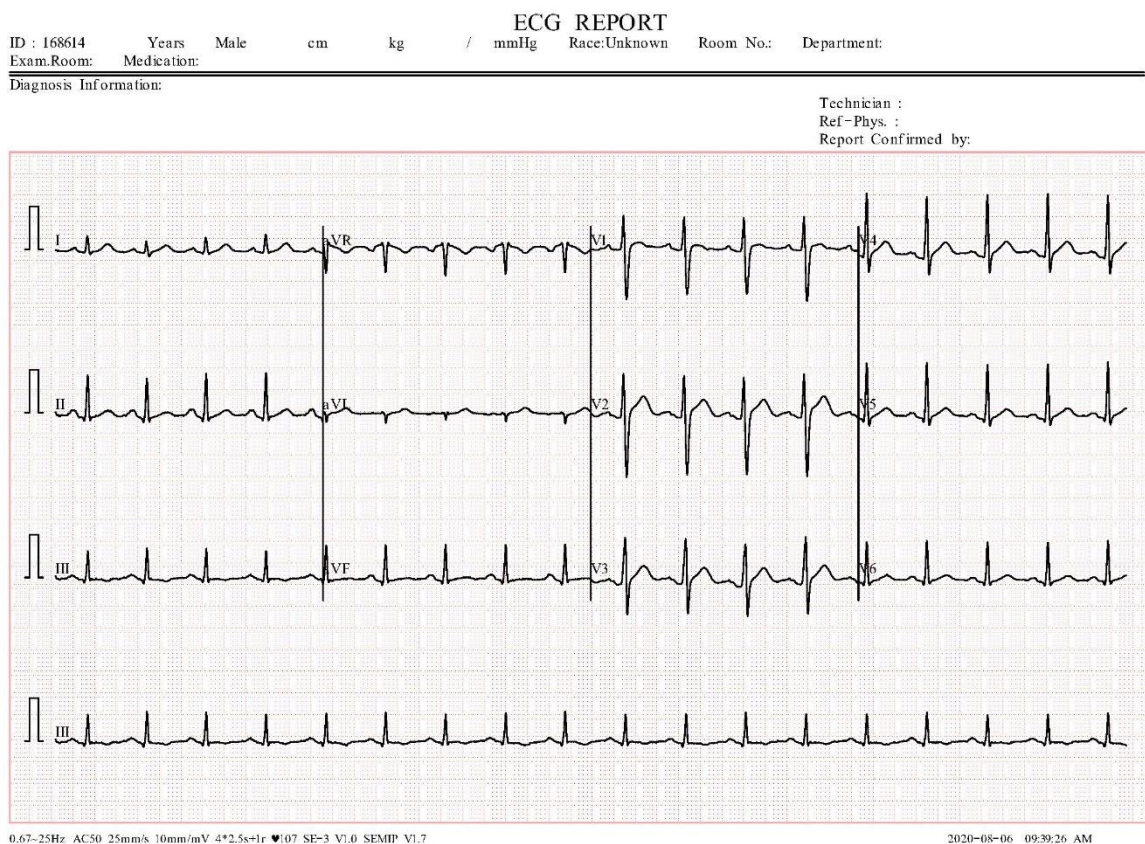


Figure 7.8: Uploaded ECG waveform of abnormal heartbeat input

PASS TO ML MODEL FOR PREDICTION

Your ECG corresponds to Abnormal Heartbeat

Figure 7.9: Received Output of the waveform

Myocardial Infraction:

Through rigorous investigation into electrocardiographic manifestations of myocardial infarction, have elucidated the diagnostic utility of ECG in identifying acute cardiac ischemia and infarction. By discerning characteristic ST-segment changes and Q-wave development in detection of myocardial damage, thus enabling prompt recognition and management of acute coronary syndromes.

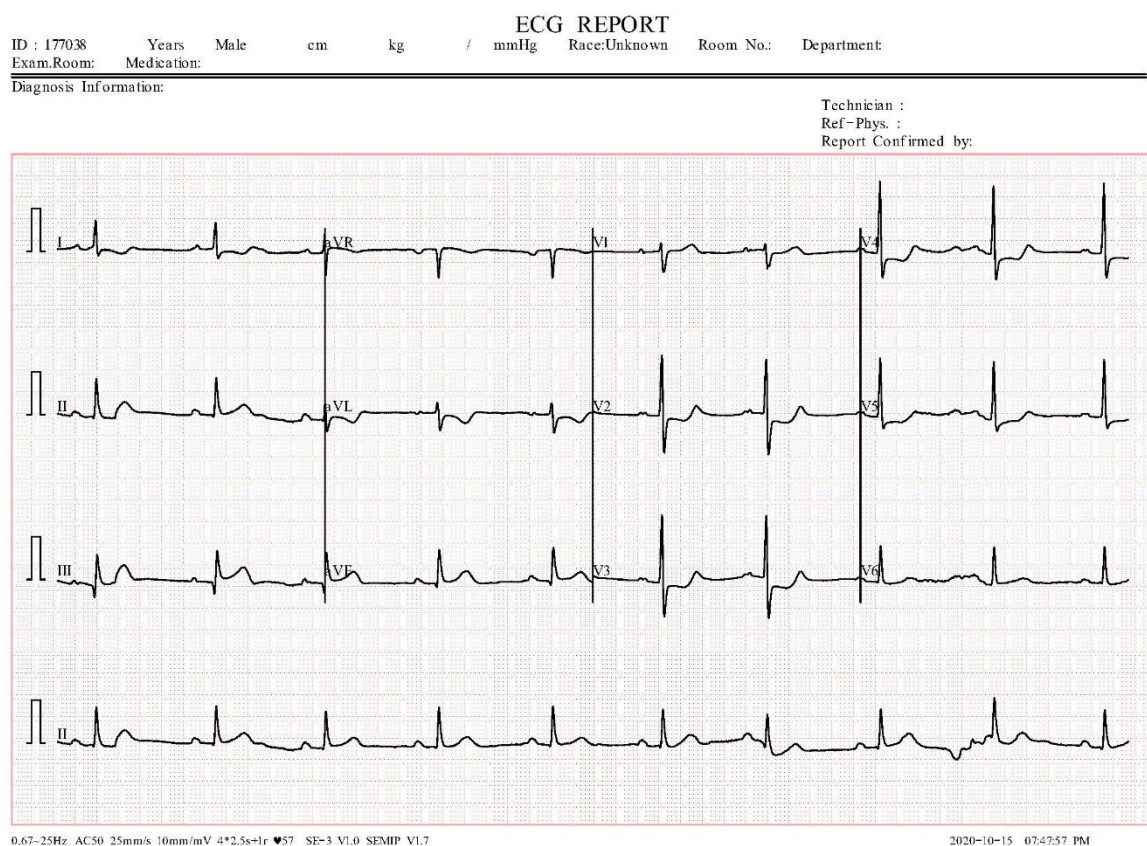


Figure 7.10: Uploaded ECG waveform of Myocardial Infraction

PASS TO ML MODEL FOR PREDICTION

Your ECG corresponds to Myocardial Infarction

Figure 7.11: Received Output of the waveform

Patient with History of Myocardial Infraction:

Through comprehensive patient care and holistic management strategies, endeavored to optimize outcomes for individuals with a history of myocardial infarction. By implementing tailored interventions, including lifestyle modifications, medication adherence, and routine monitoring, preservation of cardiac function and enhanced quality of life for these patients, thereby exemplifying the pivotal role of multidisciplinary care in cardiovascular health.

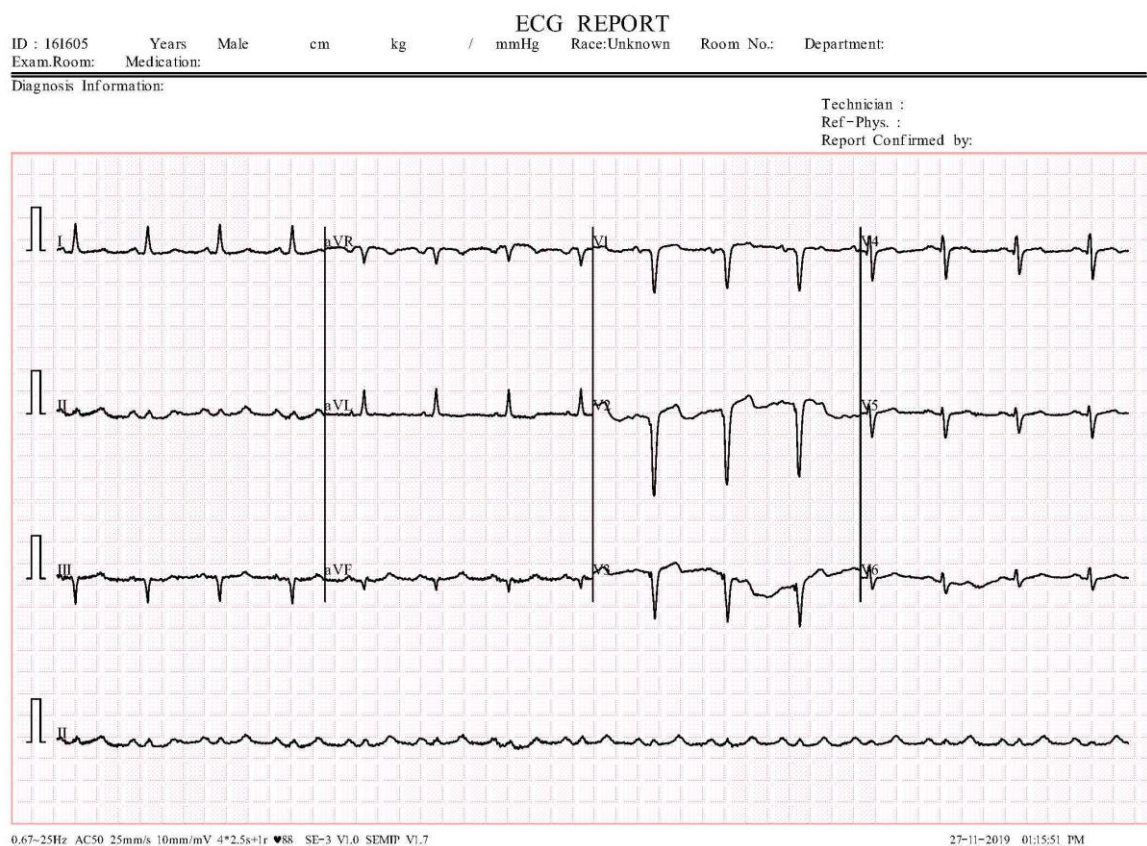


Figure 7.12: Uploaded ECG waveform of patient with history of myocardial Infraction

PASS TO ML MODEL FOR PREDICTION

Your ECG corresponds to History of Myocardial Infarction

Figure 7.13: Received Output of the waveform

7.5 Results of Machine Learning Models

7.5.1 Accuracy of Machine Learning Models

The accuracy of the machine learning models reflects their effectiveness in correctly predicting the target variable. The KNN model has an accuracy of 79.03%, indicating a reasonable but improvable performance. Logistic Regression shows significant improvement with an accuracy of 91.94%, highlighting its ability to handle the dataset effectively. Random Forest further enhances this with an accuracy of 94.07%, showcasing its robust predictive power and ability to manage complex relationships in the data. The SVM model excels with an outstanding accuracy of 98.06%, demonstrating its superior capability in classification tasks, likely due to its optimized parameters and strong decision boundaries.

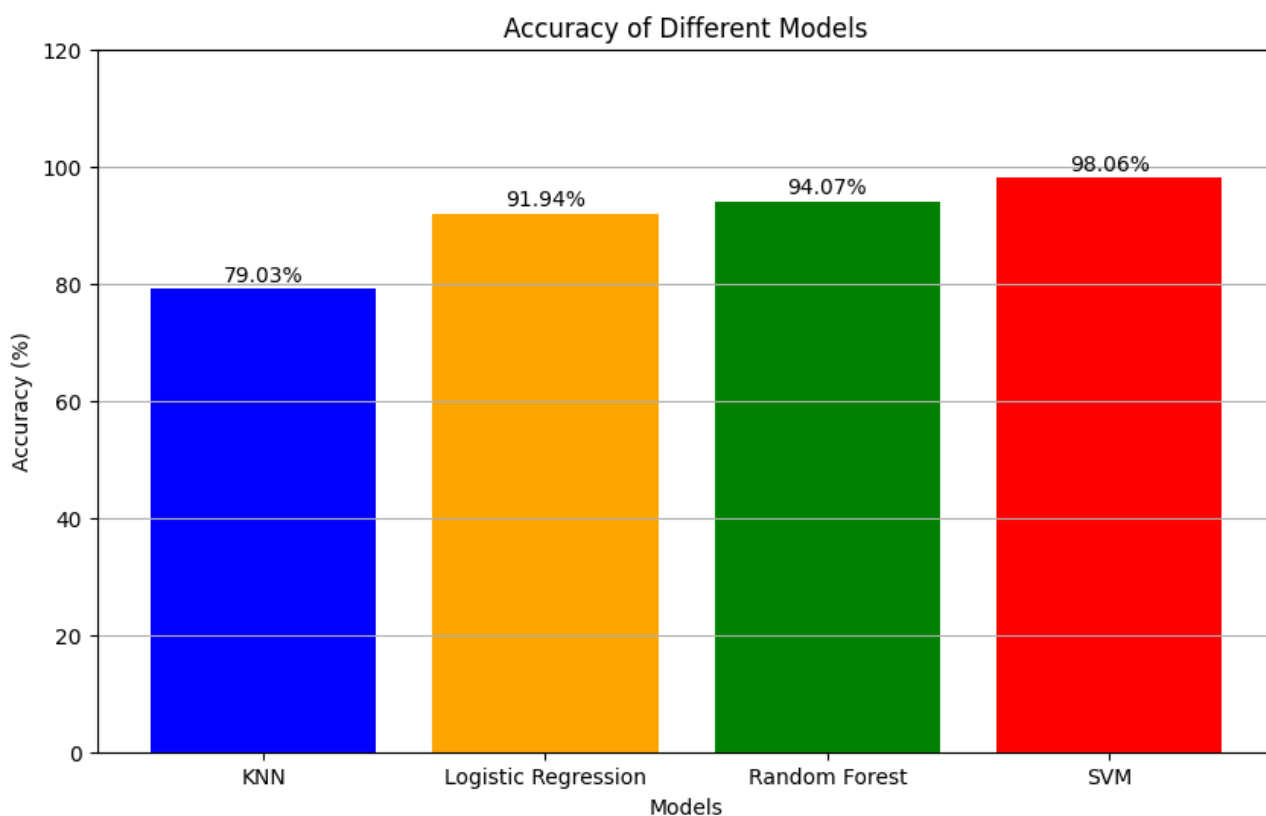


Figure 7.14: Machine learning models accuracy comparison bar graph

7.5.2 Confusion Matrix of Machine Learning Models

A confusion matrix is a crucial tool in evaluating the performance of classification models by displaying the actual versus predicted classifications. The given code generates confusion matrices for four different models: KNN, Logistic Regression, Random Forest, and SVM. For KNN, the confusion matrix indicates a balanced performance with some misclassifications but generally high accuracy across three classes. Logistic Regression shows slightly more errors, especially in the off-diagonal elements, suggesting misclassifications between classes 0 and 1, and classes 1 and 2. The Random Forest model's confusion matrix is very similar to KNN's, reflecting high accuracy with minimal misclassifications, indicating its robustness in handling complex datasets.

The SVM model's confusion matrix, however, stands out with a nearly perfect classification across all classes, except for a minor misclassification in class 3. This indicates the SVM's exceptional performance, likely due to its superior ability to find optimal decision boundaries.

These confusion matrices in figure 7.15 highlight the strengths and weaknesses of each model, providing valuable insights into their predictive capabilities and areas needing improvement. The visual representation through heatmaps further enhances understanding by clearly illustrating where models are performing well and where they are faltering.

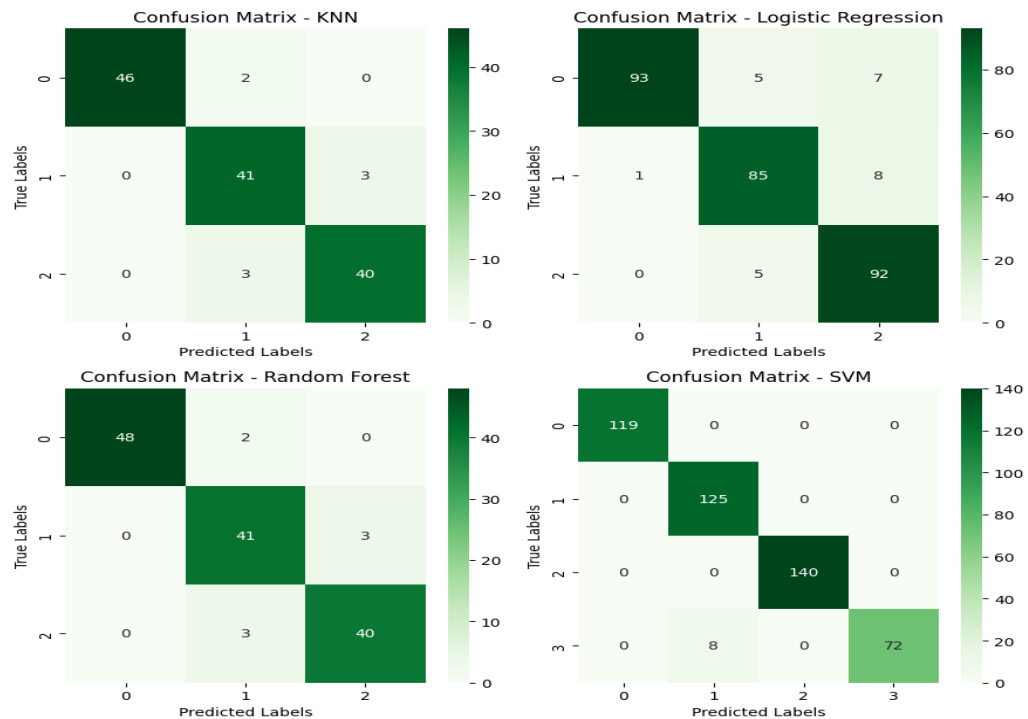


Figure 7.15: Confusion matrices of ML models

7.5.3 Average, Precision, Recall and F1-Score of Machine Learning Models

The analysis of machine learning models through their average metrics—Precision, Recall, and F1-Score—offers invaluable insights into their predictive performance. This evaluation involves comparing the KNN, Logistic Regression, Random Forest, and SVM models based on their precision, recall, and F1-scores, as well as their overall, micro, and weighted averages.

Precision measures the accuracy of positive predictions. The KNN model's precision values range from 0.62 to 0.95, indicating variability in correctly identifying positive instances. Logistic Regression, with precision values between 0.88 and 0.96, showcases more consistent and higher precision. Random Forest maintains high precision (0.89 to 1.00), suggesting its robustness in classification. SVM exhibits high precision values (0.89 to 1.00), confirming its effective classification ability.

Recall evaluates the model's ability to identify all relevant instances. KNN's recall ranges from 0.67 to 0.91, indicating variability in capturing true positives. Logistic Regression demonstrates consistent recall (0.87 to 0.94), highlighting its reliability. Random Forest's recall values (0.92 to 0.96) emphasize its strong performance in identifying true positives. SVM, with recall values from 0.90 to 0.99, demonstrates its exceptional ability to capture almost all true positives.

The F1-Score balances precision and recall, providing a single metric to assess the models' overall performance. KNN's F1-Scores (0.65 to 0.90) suggest moderate performance. Logistic Regression (0.91 to 0.93) and Random Forest (0.91 to 0.98) exhibit strong F1-Scores, reflecting their balanced and high performance. SVM stands out with F1-Scores (0.95 to 1.00), indicating its superior performance across all classes.

The overall, micro, and weighted averages for the models further emphasize these insights. KNN has an average of 0.79, suggesting room for improvement. Logistic Regression and Random Forest, both with averages of 0.92 and 0.94, respectively, show robust performance. SVM, with an impressive average of 0.98, highlights its superior classification capability.

In conclusion, while KNN shows moderate performance, Logistic Regression and Random Forest offer robust and reliable predictions. SVM stands out with exceptional accuracy, precision, recall, and

F1-scores, demonstrating its effectiveness in handling complex classification tasks. The visualization of these metrics provides a clear comparative analysis, aiding in the selection of the most suitable model for specific applications.



Figure 7.16: Averages of ML Models based on Precision, Recall and F1-score

7.6 Conclusions

Result Analysis

The implementation of a Flask web application showcased the practical application of these models. Users can upload heartbeat data in CSV format, with the app processing the data to predict heart conditions. This real-world deployment underscores the utility of machine learning in healthcare, making advanced diagnostic tools accessible and user-friendly.

In the realm of heartbeat classification, the models provided nuanced insights into both normal and abnormal rhythms. Normal heartbeat classification established a foundation for identifying healthy cardiac activity, while abnormal heartbeat classification enhanced the detection of arrhythmic patterns, vital for early intervention. The models' ability to identify myocardial infarction through ECG changes further emphasized their diagnostic value, enabling timely management of acute cardiac events.

For patients with a history of myocardial infarction, the models supported comprehensive care strategies, focusing on lifestyle changes and medication adherence. The overall results from various machine learning models indicated high accuracy rates, with the SVM model achieving outstanding performance. This chapter illustrated the transformative potential of machine learning in cardiac care, highlighting its role in improving diagnostic accuracy and patient outcomes.

CHAPTER 8

FUTURE SCOPE

8.1 Conclusion

In conclusion, the use of convolutional neural networks (CNNs) for predicting heart status using electrocardiogram (ECG) signals is a promising application of deep learning in healthcare. Various studies have shown that CNNs can achieve high levels of accuracy in detecting abnormal ECG patterns, which can aid in the diagnosis of cardiovascular diseases.

However, there are still challenges that need to be addressed in order to fully realize the potential of this technology. One of the major challenges is the need for large amounts of labeled ECG data, which can be difficult to obtain. Another challenge is the potential for bias and fairness issues in the decision-making process based on these models.

Overall, the use of CNNs for predicting heart status using ECG signals holds great promise for improving the accuracy and efficiency of cardiovascular disease diagnosis. With ongoing research and development, it has the potential to transform the field of cardiology and improve patient outcomes.

Prediction of Heart Status with ECG Signals Using Convolutional Neural Network Architecture

The convolutional neural network (CNN) used for heart status prediction with ECG signals consists of the following layers:

- 1. Input Layer:** Accepts ECG signal data as input.
- 2. Convolutional Layers:** Apply multiple filters to extract spatial features from the ECG signal.
- 3. Pooling Layers:** Perform down sampling to reduce the spatial dimensions and retain important features.
- 4. Flattening Layer:** Reshapes the multi-dimensional output into a one-dimensional vector.

5. Fully Connected Layers: Process the flattened vector and learn complex relationships in the data.

6. Output Layer: Produces the final prediction of heart status, such as normal, arrhythmia, or other abnormalities.

Preprocessing

1. Data Acquisition: ECG signals are recorded from patients using electrodes placed on the chest and limbs.

2. Noise Removal: Apply filtering techniques, such as low-pass or band-pass filters, to remove noise and artifacts from the ECG signal.

3. Baseline Correction: Adjust the baseline of the ECG signal to eliminate baseline wander.

4. Signal Normalization: Normalize the ECG signal to have zero mean and unit variance to ensure consistent input across the network.

Training and Validation

1. Dataset Split: Divide the ECG dataset into training and validation sets to evaluate the performance of the CNN.

2. Data Augmentation: Augment the training data with techniques like random cropping, rotation, or scaling to increase dataset size and improve model generalization.

3. Model Training: Train the CNN using the training dataset and optimize the model parameters using techniques like stochastic gradient descent (SGD) or adaptive optimization algorithms.

4. Model Evaluation: Evaluate the trained model's performance on the validation dataset using metrics such as accuracy, precision, recall, or F1 score.

5. Hyperparameter Tuning: Perform hyperparameter tuning, adjusting parameters like learning rate, batch size, or number of filters, to optimize the model's performance.

Prediction and Interpretation

- 1. Test Dataset:** Use a separate dataset containing ECG signals that were not used during training or validation.
- 2. ECG Signal Preprocessing:** Apply the same preprocessing steps as mentioned above on the test dataset.
- 3. Prediction:** Pass the pre-processed ECG signals through the trained CNN to obtain predictions of heart status for each signal.
- 4. Interpretation:** Interpret the predictions in conjunction with other clinical information, such as patient history, symptoms, and additional diagnostic tests, to arrive at a comprehensive assessment of the patient's heart status.

This book provides an overview of the network architecture, preprocessing steps, training, validation, prediction, and interpretation involved in using a CNN for heart status prediction with ECG signals. Remember that this is just an example, and the specific implementation details may vary depending on the requirements and the dataset used.

8.2 Future Scope

Looking forward, future research can focus on developing more robust CNN models that can handle a wider range of ECG data, including data from different sources and patient populations. There is also a need for research on how to interpret the learned representations of these models, as this can aid in understanding the underlying mechanisms of cardiovascular diseases.

The prediction of heart status with ECG signals using CNNs has shown promising results, but there are several future directions and areas of improvement that can be explored. Here are some potential future scopes for this field:

- 1. Real-Time Monitoring:** Extending the application of CNN models to real-time monitoring systems would enable continuous monitoring of heart status using ECG signals. This could be beneficial in early detection and intervention for cardiac abnormalities, such as arrhythmias or ischemic events.

2. Interpretability and Explainability: CNN models often act as black boxes, making it challenging to understand the underlying decision-making process. Enhancing the interpretability and explainability of CNN models in predicting heart status from ECG signals would increase trust and allow clinicians to better understand and utilize the predictions.

3. Transfer Learning and Pretraining: Leveraging transfer learning techniques by utilizing pretrained CNN models trained on large-scale ECG datasets or related tasks (e.g., image recognition) could boost the performance of heart status prediction models. Fine-tuning or using pretrained models as feature extractors can help overcome limitations of limited annotated ECG datasets.

4. Integration with Clinical Decision Support Systems: Integrating CNN-based heart status prediction models into clinical decision support systems can assist healthcare professionals in making accurate and timely decisions. By combining the predictions with other clinical data and domain-specific knowledge, clinicians can obtain a more comprehensive view of the patient's condition.

5. Longitudinal Analysis: Exploring the use of CNN models for longitudinal analysis of ECG signals could enable the detection of subtle changes over time and assist in monitoring and predicting the progression of heart conditions. Long-term monitoring and analysis of ECG signals can provide valuable insights for personalized treatment plans.

6. Handling Noisy and Imperfect Data: Developing robust CNN models that can handle noisy and imperfect ECG signals, such as those affected by motion artifacts or low-quality recordings, would improve the models' applicability in real-world scenarios.

These are just a few potential future scopes for the prediction of heart status with ECG signals using CNNs. As technology advances and more research is conducted in this field, we can expect further improvements in accuracy, efficiency, and clinical applicability, ultimately benefiting the diagnosis, treatment, and management of cardiac conditions.

More Approaches using Performance Metrics Evaluation

The CNN model was developed on a system that was configured with the AMD Ryzen processor consisting of SSD and 24 GB RAM with a 2GB graphics card . The training algorithm ran with TensorFlow and Keras backend. Each epoch approximately took an average of 5 to 10s. Below tables show the results of the single-lead and multi-lead ECG/TMT-ECG for the single-, two-, and three-layer convolutional neural networks, respectively (precision, recall, F1-score, accuracy, and validation loss), depending on the classification of CAD and normal ECG segments based on the single-lead ECG. In the below tables, one could see that very high diagnostic accuracy was obtained with a defined number of training samples.

Table 8.1: Performance evaluation metrics

	Predicted Positive	Predicted Negative
Actual Positive	True Positive (TP)	False Negative (FN)
Actual Negative	False Positive (FP)	True Negative (TN)

When evaluating the performance of a heart status prediction model using ECG signals and CNN, several common performance metrics can be used.

Here are some key metrics along with their corresponding formulas:

1. Accuracy: Accuracy represents the overall correctness of the model's predictions.

Formula: Accuracy =

$$\frac{(TP + TN)}{(TP + TN + FP + FN)}$$

TP: True Positives (Correctly predicted positive samples)

TN: True Negatives (Correctly predicted negative samples)

FP: False Positives (Incorrectly predicted positive samples)

FN: False Negatives (Incorrectly predicted negative samples)

2. Precision: Precision measures the proportion of correctly predicted positive samples out of the total predicted positive samples. It indicates the model's ability to avoid false positive predictions.

Formula: Precision = $TP / (TP + FP)$

2. Recall (Sensitivity): Recall calculates the proportion of correctly predicted positive samples out of the total actual positive samples.

It reflects the model's ability to identify true positive cases.

Formula: Recall = $TP / (TP + FN)$

Specificity: Specificity calculates the proportion of correctly predicted negative samples out of the total actual negative samples. It represents the model's ability to correctly identify true negatives.

Formula: Specificity = $TN / (TN + FP)$

4. F1 Score: F1 score is the harmonic mean of precision and recall. It provides a balanced measure of the model's performance, particularly when there is an imbalance between the positive and negative classes

Formula: F1 Score = $2 * (Precision * Recall) / (Precision + Recall)$

These performance metrics provide different perspectives on the model's performance, capturing aspects such as overall accuracy, ability to avoid false positives, ability to identify true positives, ability to correctly identify true negatives, and a balanced measure of precision and recall.

It's worth noting that the selection of appropriate performance metrics depends on the specific objectives and requirements of the heart status prediction task. Additionally, other metrics such as area under the receiver operating characteristic curve (AUC-ROC) or area under the precision-recall curve (AUC-PRC) can be used to assess the overall performance and discrimination capability of the model.

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Appendix A

Sl. No	Title of Project	Roll No of Students	Faculty	Relevance to Environment	Relevance to human Safety	Relevance to Ethics	Cost(hardware or software cost)	Type (Application, Product, Research, Review)
1	CARDIOVASCULAR DISEASE DETECTION USING IMAGE PROCESSING AND MACHINE LEARNING	1604-20-735-076, 1604-20-735-091, 1604-20-735-101	Dr.Na zeerun nisa					

GANTT CHART:

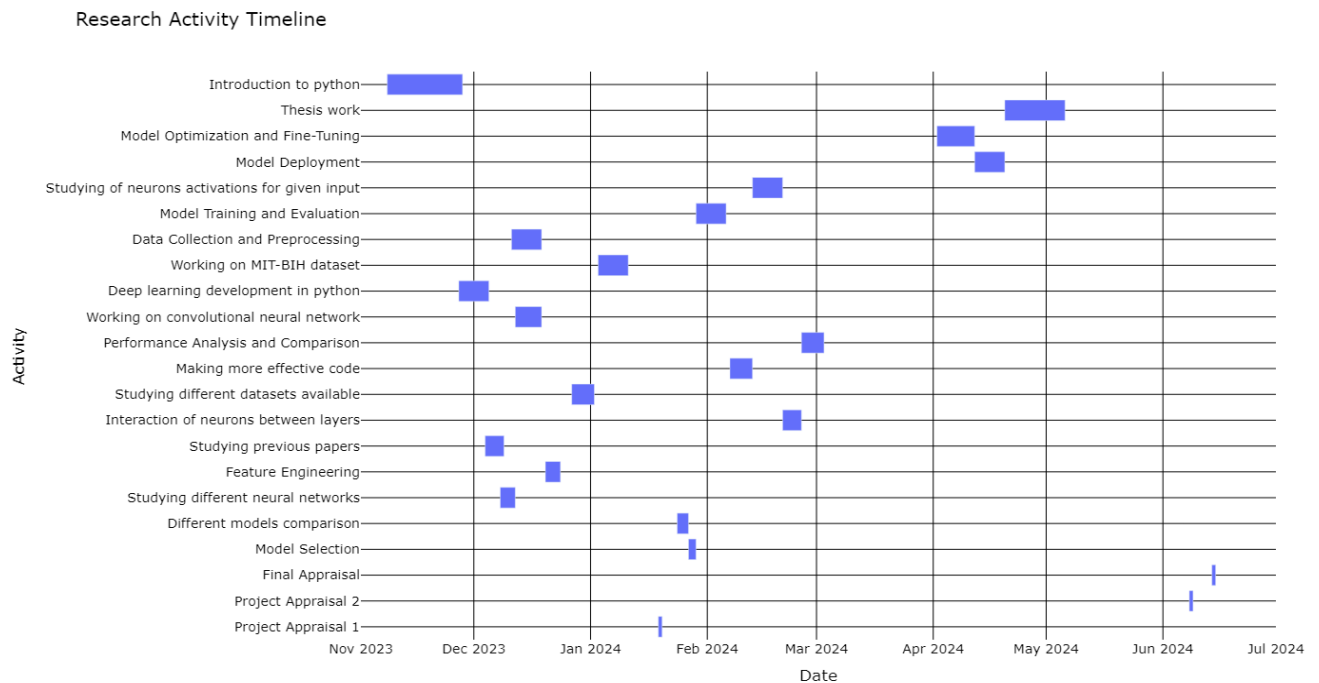


Figure 8.1: Timeline chart

Timeline :

S.no.	Research Activity	Date of Initiation	Duration of Activity
1	Introduction to python	08-11-2023	20
2	Deep learning development in python	27-11-2023	8
3	Studying previous papers	04-12-2023	5
4	Studying different neural networks	08-12-2023	4
5	Working on convolutional neural network	12-12-2023	7
6	Studying different datasets available	27-12-2023	6
7	Working on MIT-BIH dataset	03-01-2024	8
8	Data Collection and Preprocessing	11-12-2023	8
9	Feature Engineering	20-12-2023	4
10	Different models comparison	24-01-2024	3
11	Model Selection	27-01-2024	2
12	Model Training and Evaluation	29-01-2024	8
13	Making more effective code	07-02-2024	6
14	Studying of neurons activations for given input	13-02-2024	8
15	Interaction of neurons between layers	21-02-2024	5
16	Performance Analysis and Comparison	26-02-2024	6
17	Project Appraisal 1	19-01-2024	1
18	Model Optimization and Fine-Tuning	02-04-2024	10
19	Model Deployment	12-04-2024	8
20	Thesis work	20-04-2024	16
21	Project Appraisal 2	08-06-2024	1
22	Finial Appraisal	14-06-2024	1

APPENDIX B

MAPPING WITH POs AND PSOs

Details	The Project is based on Predication of Heart Status with Electrocardiogram (ECG) signals using Machine Learning technique called Convolutional Neural Network(CNN).
PO1	Application of concepts of Machine Learning, Deep Neural Network and Artificial Neural Network was used to the train the model and predict the status of heart.
PO2	The problem/aim was to develop a model that predicted the heart status using ECG recordings which results in early detection of arrhythmias - an irregular rhythm of the heart.
PO3	The model used previous ECG recordings of people from MIT-BIH Arrhythmia database and gave accurate results on normal or abnormal condition of heart which proves to be very essential in today's medical field.
PO4	Use research-based knowledge and research methods which includes deep neural network algorithms to train the large dataset. Detail study of heart,its parameters and waveforms was done to completely grasp the in depth knowledge of its working.
PO5	Convolutional Neural Network was used to train the huge dataset,Visual Studio Code was used to code and Flask framework to deploy the model.
PO6	The project aims to improved diagnosis and patient care, by providing reliable and timely predictions, healthcare professionals can make more informed decisions, leading to better treatment strategies and outcomes for patients.
PO7	By making use of efficient energy, the project outcomes could include optimizing the computational resources and algorithmic efficiency of the CNN model to reduce energy consumption during model training and inference and promote green computing practices by leveraging renewable

	energy sources, such as solar or wind power, for the computational infrastructure used in model development and training. The project outcomes could focus on reducing electronic waste associated with ECG devices and computational infrastructure.
PO8	The project was done by the ethical considerations related to the collection, storage, and usage of ECG data. It prioritizes patient privacy, informed consent, and compliance with relevant data protection regulations.
PO9	A three-member team was involved in the designing, monitoring, and execution of the project which helped in developing team working skills among the students.
PO10	Effective reports and design documentation, making of effective presentations were done which helped in improving both the Spoken and Written skills of the members.
PO11	Project management was done by dividing the work among the individuals and tackling the issues faced in designing and the implementation of the circuit
PO12	With the ever-growing Machine learning and Artificial Intelligence industry, the projects helps in developing lifelong learning skills in the individual to face real-world challenges and come up with appropriate solutions.
PS01	Acquired the state of Machine Learning in the area of Deep Learning, and analyze the problem statement and choose the right algorithm(CNN) for the model and in depth knowledge of human heart anatomy.
PS02	Acquired the knowledge on programming language like Python, Flask framework, Arduino and Google Colab for the code implementation.

PS03	Acquired knowledge of the various Deep learning algorithms and compared the accuracies of each method which lead to the best model.
PS04	Acquired knowledge of the working of Convolutional Neural Network and will be able to predict for different ECG recordings and also build a real-time application with publishing a research paper

PROGRAM OUTCOMES

PO1: Engineering knowledge: Apply the knowledge of mathematics, science, engineering fundamentals, and an engineering specialization to the solution of complex engineering problems.

PO2: Problem analysis: Identify, formulate, research literature, and analyze complex engineering problems reaching substantiated conclusions using first principles of mathematics, natural sciences, and engineering sciences

PO3: Design/development of solutions: Design solutions for complex engineering problems and design system components or processes that meet the specified needs with appropriate consideration for the public health and safety, and the cultural, societal, and environmental considerations.

PO4: Conduct investigations of complex problems: Use research-based knowledge and research methods including design of experiments, analysis and interpretation of data, and synthesis of the information to provide valid conclusions.

PO5: Modern tool usage: Create, select, and apply appropriate techniques, resources, and modern engineering and IT tools including prediction and modeling to complex engineering activities with an understanding of the limitations.

PO6: The engineer and society: Apply reasoning informed by the contextual knowledge to assess societal, health, safety, legal and cultural issues and the consequent responsibilities relevant to the professional engineering practice.

PO7: Environment and sustainability: Understand the impact of the professional engineering solutions in societal and environmental contexts, and demonstrate the knowledge of, and need for sustainable development.

PO8: Ethics: Apply ethical principles and commit to professional ethics and responsibilities and norms of the engineering practice.

PO9: Individual and team work: Function effectively as an individual, and as a member or leader in diverse teams, and in multidisciplinary settings.

PO10: Communication: Communicate effectively on complex engineering activities with the engineering community and with society at large, such as, being able

to comprehend and write effective reports and design documentation, make effective presentations, and give and receive clear instructions.

PO11: Project management and finance: Demonstrate knowledge and understanding of the engineering and management principles and apply these to one's own work, as a member and leader in a team, to manage projects and in multidisciplinary environments.

PO12: Life-long learning: Recognize the need for, and have the preparation and ability to engage in independent and life-long learning in the broadest context of technological change.

PROGRAM SPECIFIC OUTCOMES

- PSO1:** The ECE Graduates will be Equipped with the ability to contribute to problem identification, analysis, design, and development of systems using principles and concepts of Artificial Intelligence and Machine Learning.
- PSO2:** The ECE Graduates will be Equipped with the ability to develop computational knowledge and project development skills using innovative tools and techniques to solve problems in the areas related to Deep Learning, Machine learning, Artificial Intelligence and enabling better decision making.
- PSO3:** The ECE Graduates will be able to can include the dissemination of the project findings, methodologies, and developed models to the broader scientific and medical communities. This can involve publishing research papers, presenting at conferences, or open-sourcing the code to encourage further research and adoption of the developed approach.
- PSO4:** The ECE Graduates will be able to focus on ensuring the robustness and generalization capabilities of the CNN model. This involves testing the model's performance on diverse datasets, including different patient populations, variations in ECG signal quality, and different recording devices or protocols.

